PHYSICS

1) In YDSE, Bichromatic light of wavelength λ_1 = 6000Å and λ_2 = 7000Å is used. Find for bright fringe minimum order of λ_1 which overlapped with λ_2

(1) 7th bright fringe of λ_1 is overlapped with 6th bright fringe of λ_2.

(2) 4th bright fringe of λ_1 is overlapped with 3rd bright fringe of λ_2.

(3) 4th bright fringe of λ_1 is overlapped with 2nd bright fringe of λ_2.

(4) None of these

2) The angle of polarisation for a medium is 60° what will be the criticle angle for this medium at air interface :-

(1) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(2) $\sin^{-1}\left(\sqrt{3}\right)$

(3) $\cos^{-1}\left(\sqrt{3}\right)$

(4) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

3) In YDSE, one of the slit is wider than other, so that amplitude of the light from one slit is double of that from other slit. If I_m be the maximum intensity, the resultant intensity I when they interfere at a phase difference ϕ is given by :-

(1) $\frac{I_m}{9}(4 + 5\cos\phi)$

(2) $\frac{I_m}{3}[1 + 2\cos^2(\phi/2)]$

(3) $\frac{I_m}{5}[1 + 4\cos^2(\phi/2)]$

(4) $\frac{I_m}{9}[1 + 8\cos^2(\phi/2)]$

4) **Assertion (A) :-** In Young's double slit experiment the two slits are at distance d apart. Interference pattern is observed on a screen at fixed distance D from the slits. At a point on the screen when it is directly opposite to one of the slits, a dark fringe is observed. Then, the wavelength of wave is proportional to square of distance of two slits. **Reason (R) :-** For a dark fringe in YDSE intensity is zero.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

5) In single slit diffraction, the phase difference between the Huygen's wavelet from the edge of the slit and the wavelet from the mid point of the slit for first maxima.

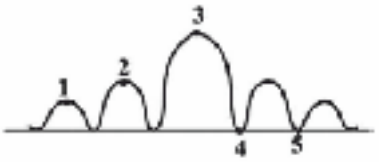
(1) 2π radian

(2) π radian

(3) $\frac{5\pi}{2}$ radian

(4) $\frac{3\pi}{2}$ radian

6) The diffraction pattern of a single slit is shown in the figure. The point at which the path difference of the extreme rays is two times the wavelength is



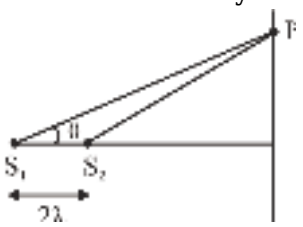
(1) point 1

(2) point 2

(3) point 4

(4) point 5

7) In young's double slit experiment, the slits are horizontal. The intensity at a point P as shown in figure is $\frac{3}{4}I_0$, where I^0 is the maximum intensity. Then the value of θ is (given the distance between the two slits S_1 and S_2 is 2λ)



(1) $\cos^{-1}\left(\frac{1}{12}\right)$

(2) $\sin^{-1}\left(\frac{1}{12}\right)$

(3) $\tan^{-1}\left(\frac{1}{12}\right)$

(4) $\sin^{-1}\left(\frac{3}{5}\right)$

8) Nuclear forces are :-

(1) Short ranged, attractive and charge independent.

(2) Short ranged, attractive and charge dependent.

(3) Long ranged, repulsive and charge independent.

(4) Long ranged, repulsive and charge dependent.

9) If energy released per fission is 200 MeV, then the number of fission reactions of ${}_{92}U^{235}$ per second for the production of 1 kW power will be-

(1) 3.5×10^{10}

(2) 4.2×10^{11}

(3) 3.125×10^{12}

(4) 3.125×10^{13}

10) Given below are two statements :
Statement I : Most of the mass of the atom and all its positive charge are concentrated in a tiny nucleus and the electrons revolve around it, is Rutherford's model.
Statement II : An atom is a spherical cloud of positive charges with electrons embedded in it, is a special case of Rutherford's model.
In the light of the above statements, choose the most appropriate from the options given below.

(1) Both statement I and statement II are false

(2) Statement I is false but statement II is true

(3) Statement I is true but statement II is false

(4) Both statement I and statement II are true

11) The ratio of wavelength of first line of lyman series of doubly ionized lithium atom to that of the first line of lyman series of deuterium (${}_1H^2$) will be

(1) 4 : 1

(2) 1 : 4

(3) 9 : 1

(4) 1 : 9

12) In the given reaction ${}_ZX^A \rightarrow {}_{Z'}Y^{A'} + n\alpha + n'\beta^o$. The ratio $\left(\frac{Z'}{A'}\right)$ will be equal to :-

(1) $\frac{Z + n' + 2n}{A - 4n}$

(2) $\frac{Z + n' - 2n}{A - 4n}$

(3) $\frac{Z - n' + 2n}{A + 4n}$

(4) $\frac{Z + n' - 2n}{A + 4n}$

13)

A monochromatic source of light operating at 200W emits 4×10^{20} photons per second then (λ) wavelength of light is →

(1) 200 nm

(2) 400 nm

(3) 100 nm

(4) 1800 nm

14) If 5% of the energy supplied to a bulb is irradiated as visible light, how many photon are emitted per second by a 100 W lamp (Assume wavelength of visible light as 5.6×10^{-5} cm) :-

(1) 1.4×10^{19}

(2) 3×10^{3}

(3) 1.4×10^{-18}

(4) 3×10^{4}

15) Wave equation of electric field at a point is

$E = 100 \frac{V}{m}$ [sin7ωt + cos10ωt + cos15ωt] at 't' instant. If work function of photo cell is ϕ, then stopping potential is :

(1) $\frac{h}{e}\left(\frac{16\omega}{\pi}\right) - \frac{\phi}{e}$

(2) $\frac{h}{e}\left(\frac{15\omega}{2\pi}\right) - \frac{\phi}{e}$

(3) $\frac{h}{e}\left(\frac{15\omega}{\pi}\right) - \frac{\phi}{e}$

(4) $\frac{h}{e}\left(\frac{7\omega}{2\pi} - \frac{\phi}{e}\right)$

16) All electron ejected from a surface by incident of wavelength 200 nm can be stopped before travelling 1 meter in the direction of a uniform electric field of 4 NC^{-1}. The work function of surface is :-

(1) 4 eV

(2) 6.2 eV

(3) 2 eV

(4) 2.2 eV

17) In an electron microscope, the resolution that can be achieved is of the order of wavelength of electrons used. To resolve a width of 7.5×10^{-12}m the minimum electron energy required is close to :-

(1) 100 Kev

(2) 500 Kev

(3) 25 Kev

(4) 1 Kev

18) If alpha particle and deutron move with velocity v and 2v respectively, the ratio of their de-

Broglie wave length will be :-

(1) $1 : \sqrt{2}$

(2) 2 : 1

(3) 1 : 1

(4) $\sqrt{2} : 1$

19) If the value of distance is 2m between source and cathode, the frequency of incident radiation is 10^{14} Hz then the photons per second is x now the distance becomes 4m and the frequency increases to 2×10^{14} Hz and power of source remain same then the value of photons per second (x') becomes :-

(1) $x' = \frac{x}{8}$

(2) $x' = \frac{x}{4}$

(3) x' = 2x

(4) x' = 4x

20) When an object is kept at a distance of 5cm from the pole of a concave mirror, a virtual image is formed. If the object distance is 15 cm, a real image is formed. Both the images have same magnification. The focal length of the mirror is-

(1) –10 cm

(2) –5 cm

(3) –20 cm

(4) –15 cm

21) An air bubble is situated in glass slab of length 25 cm. When it is seen from one side, it appears at 6 cm and when see from other side it appears at 4 cm. Find refractive index of slab :-

(1) 1.5

(2) 2.5

(3) 1

(4) 1.33

22) **Assertion :** The twinkling of stars is due to the fact that refractive index of the earth's atmosphere fluctuates.
Reason : In cold countries, the phenomenon of looming (i.e ship appears in the sky) takes place, because refractive index of air decreases with height.

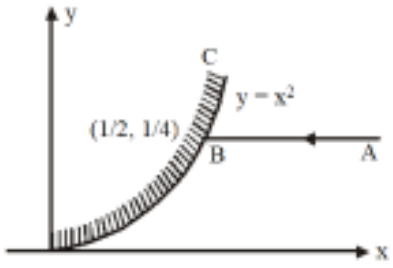
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

23) In the given figure, incident ray AB is parallel to x-axis. The angle of reflection is :-



(1) 30°

(2) 60°

(3) 45°

(4) None of these

24) When an object is at a distance x and y from the optical centre of a lens, a real and a virtual images are formed, respectively with same magnification. The focal length of lens is–

(1) x + y

(2) $\frac{x+y}{2}$

(3) $\frac{x+y}{4}$

(4) $\sqrt{xy}$

25) The focal length of objective and eyepiece of a compound microscope are 1 cm and 5 cm respectively. An object is placed at a distance of 1.1 cm from objective. If final image is formed at infinity, the distance between lenses is given by:

(1) 6 cm

(2) 10 cm

(3) 16 cm

(4) 16.7 cm

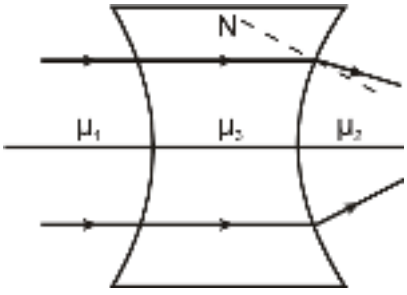
26) A ray of light making an angle 10° with the horizontal is incident on a plane mirror making an angle θ with the horizontal. What should be the value of θ so that the reflected ray goes vertically upwards ?

(1) 20°

(2) 30°

(3) 40°

(4) 45°



27) From the figure shown establish a relation between, μ_1, μ_2, μ_3 :-

(1) $\mu_3 > \mu_2 > \mu_1$

(2) $\mu_3 < \mu_2 < \mu_1$

(3) $\mu_2 > \mu_3$; $\mu_3 = \mu_1$

(4) $\mu_2 > \mu_1$; $\mu_3 = \mu_2$

28) A convergent beam of light converges to a point 20 cm behind the convex mirror on the principal axis. Its image is coincident with the virtual object. Then, the focal length of the convex mirror is :

(1) 20 cm

(2) 10 cm

(3) 40 cm

(4) 30 cm

29) An effective power of a combination of 5 identical convex lenses which are kept in contact along the principal axis is 25 D. Focal length of each of the convex lens is :

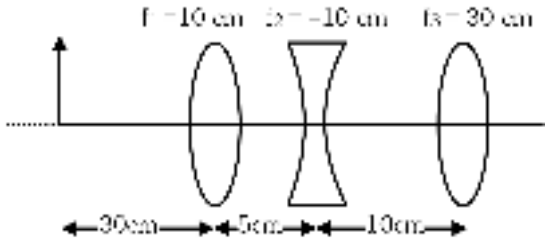
(1) 20 cm

(2) 50 cm

(3) 500 cm

(4) 25 cm

30) The position of the image formed by the combination of lenses is :



(1) 30 cm (right of third lens)

(2) 15 cm (left of second lens)

(3) 30 cm (left of third lens)

(4) 15 cm (right of second lens)

31) Given below are two statements :
Statement I : When the white light passed through a prism, the red light bends lesser than yellow and violet.
Statement II : The refractive indices are different for different wavelengths in dispersive medium.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are true.

(2) Statement I is true but Statement II is false.

(3) Both Statement I and Statement II are false.

(4) Statement I is false but Statement II is true.

32) Time taken by light to travel in two different materials A and B of refractive indices μ_A and μ_B of

same thickness is t_1 and t_2 respectively. If $t_2 - t_1 = 5 \times 10^{-10}$ s and the ratio of μ_A to μ_B is 1 : 2. Then the thickness of material, in meter is: (Given v_A and v_B are velocities of light in A and B materials respectively in m/s).

(1) $5 \times 10^{-10}\ v_A$ m

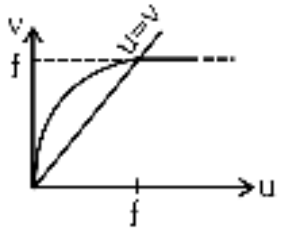
(2) 5×10^{-10} m

(3) 1.5×10^{-10} m

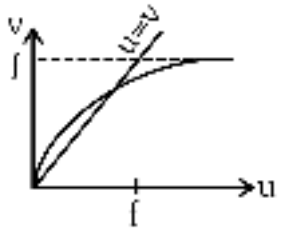
(4) $5 \times 10^{-10}\ v_B$ m

33) For a concave lens of focal length f, the relation between object and image distance u and v, respectively, from its pole can best be represented by (u = v is the reference line) :

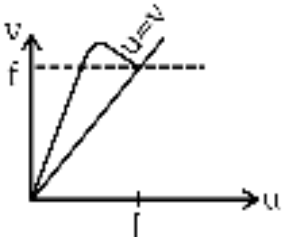
(1)



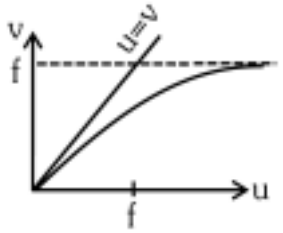
(2)



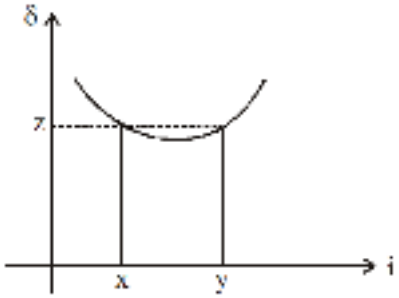
(3)



(4)



34) For a prism, graph between angle of deviation and angle of incidence is given. The angle of prism is:-



(1) x + y – z

(2) x – y + z

(3) x – y – z

(4) – x + y + z

35) Which of the given statements are correct regarding unbiased P-N Junction?
I. Drift and diffusion currents occur p to n side.
II. Initially, diffusion current is large and drift current is small.
III. Finally, diffusion and drift currents are equal in magnitude.
IV. Under equilibrium there is no net current across p-n Junction plane.

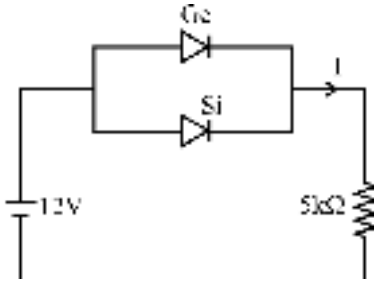
(1) I and IV

(2) I, II and III

(3) II, III and IV

(4) All of these

36) In the given circuit if Si diode and Ge diode starts conducting at 0.7 V and 0.3 V respectively then value of current I is :



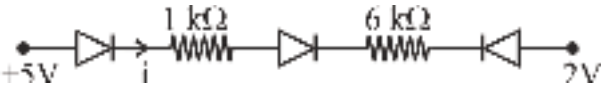
(1) 10.86 mA

(2) 4.26 mA

(3) 2.34 mA

(4) 6.5 mA

37)



Find the value of current i.

(1) 1mA

(2) 7mA

(3) zero

(4) 5mA

38) Zener breakdown in a semiconductor diode occurs when :-

(1) forward current exceeds certain value

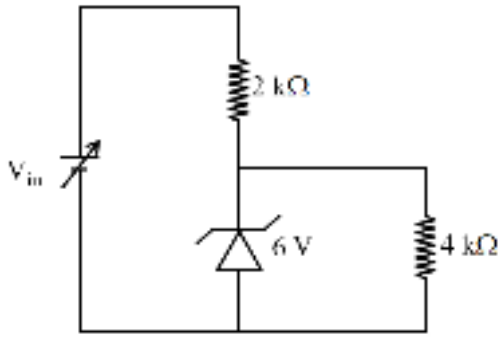
(2) Reverse biases voltage exceeds certain value

(3) forward bias voltage exceeds certain value

(4) potential barrier is reduce to zero

39) If the unregulated input voltage varies between 10 V to 16 V then what is the maximum Zener

current :-



(1) 2.5 mA

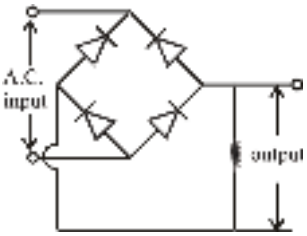
(2) 3.5 mA

(3) 7.5 mA

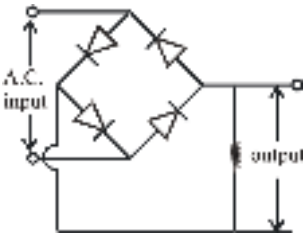
(4) 1.5 mA

40) Which of the following circuits provides full-wave rectification of A.C. input ?

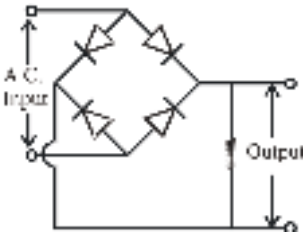
(1)



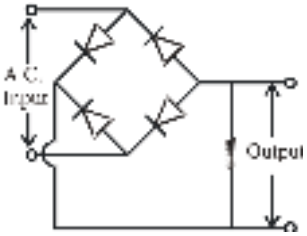
(2)



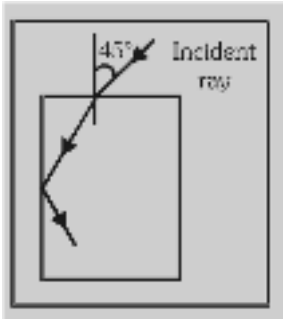
(3)



(4)



41) For the given incident ray as shown in figure, the condition of total internal reflection of this ray



the minimum refractive index of the transparent slab will be :–

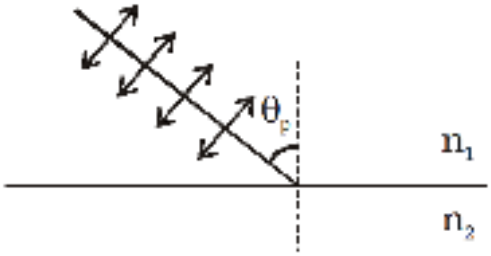
(1) $\frac{\sqrt{3}+1}{2}$

(2) $\frac{\sqrt{2}+1}{2}$

(3) $\sqrt{\frac{3}{2}}$

(4) $\sqrt{\frac{7}{6}}$

42) A plane polarized wave is incident on a boundary separating two media at brewster's angle. The plane of vibration of electric field is same as the plane of incidence. Then :-



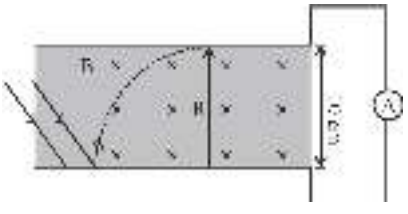
(1) There is no refracted ray

(2) There no reflected ray.

(3) Reflected ray and refracted ray are both partially polarized.

(4) Reflected ray and refracted ray are polarized in a plane perpendicular to the plane of incidence.

43) In an arrangement on photoelectric effect, the emitter and the collector plates are placed at a separation of 10cm and are connected through an ammeter without any cell. A magnetic field B exists parallel to the plates. The work function of the emitter is 2.39 eV and the light incident on it has wavelengths between 400 nm and 600 nm. Find the minimum value of B for which the current registered by ammeter is zero. Neglect any effect of space charge :-



(1) 2.85×10^{-5} T

(2) 4.90×10^{5} T

(3) 5.70×10^{3} T

(4) 4.90×10^{9} T

44) Conductivity of a photodiode starts changing only if the wavelength of incident light is less than 660 nm. The band gap of photodiode is found to be $\left(\frac{X}{8}\right)$ eV. The value of X is :
(Given, h = 6.6×10^{-34} Js, e = 1.6×10^{-19}C)

(1) 15

(2) 11

(3) 13

(4) 21

45) In experiment to find out focal length of a concave mirror by parallax method following arrangements were done on optical bench.
(i) The mirror is fixed at mark 25.0 cm
(ii) Object needle is placed at mark 55.0 cm
(iii) For no parallax between object and image, the image needle is placed at mark 85.0 cm
The estimated value of focal length is :-

(1) –10.0 cm ± 0.2 cm

(2) –20.0 cm ± 0.1 cm

(3) –20.0 cm ± 0.5 cm

(4) –10.0 cm ± 0.1 cm

CHEMISTRY

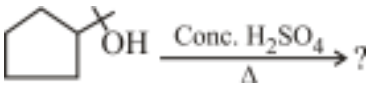
1) What is the final product of given reaction ?
$CH_3—C\equiv CH \xrightarrow{Na} \xrightarrow{CH_3CH_2I}$ product

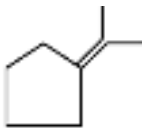
(1) $CH_2 = CH—CH_2CH_3$

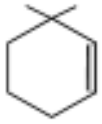
(2) $CH_3—CH_2—C\equiv C—CH_2—CH_3$

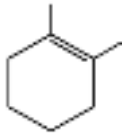
(3) $CH\equiv C—CH_2—CH_2CH_3$

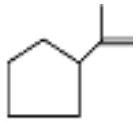
(4) $CH_3—C\equiv C—CH_2CH_3$

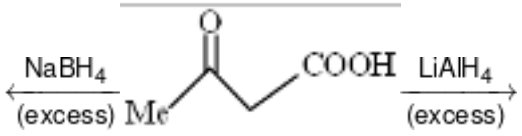
2)  Final product is:-

(1) 

(2) 

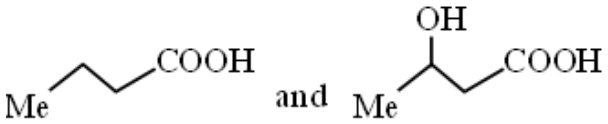
(3) 

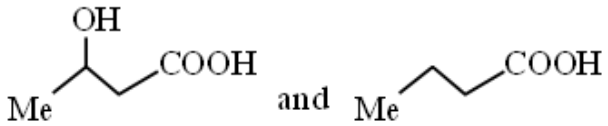
(4) 

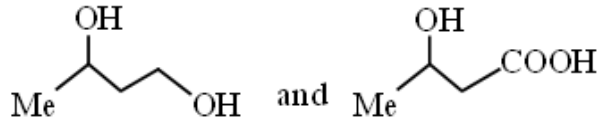
3) (B) 

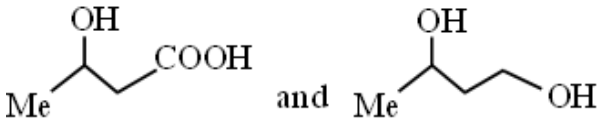
(A)

A and B respectively are :

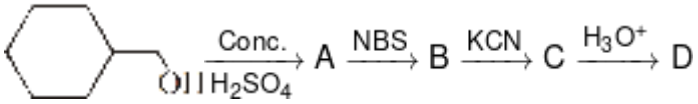
(1) 

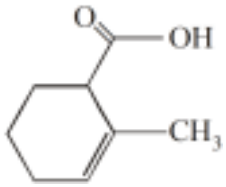
(2) 

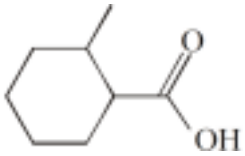
(3) 

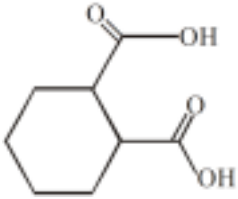
(4) 

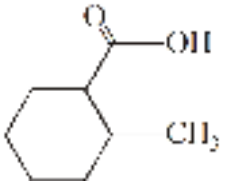
4) Identify product D of the below mentioned reaction :

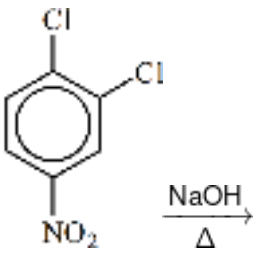


(1) 

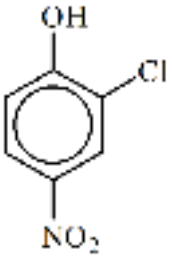
(2) 

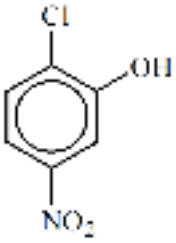
(3) 

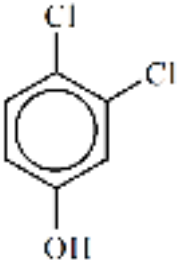
(4) 

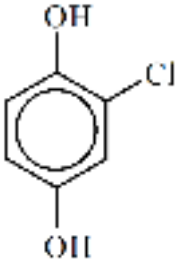
5) 

Product of given reaction is :-

(1) 

(2) 

(3) 

(4) 

6) Which of the following will form a yellow precipitate of iodoform on heating with I_2 and dilute NaOH.

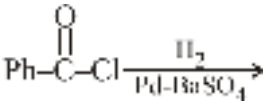
(1) $PhCH_2CH_2OH$

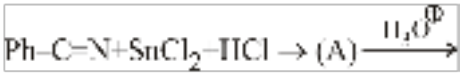
(2) $PhCH_2CHOHCH_2CH_3$

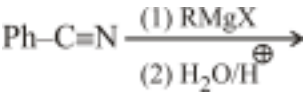
(3) $CH_3CH_2CH_2OH$

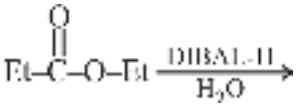
(4) $CH_3CHOHCH_3$

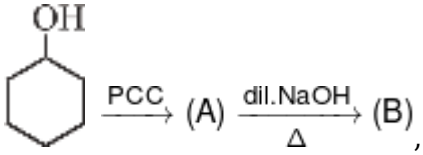
7) In which of the following reaction one of the final product is not aldehyde

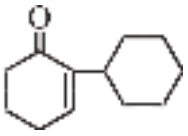
(1) 

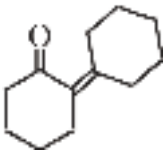
(2) 

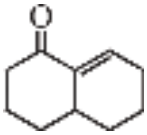
(3) 

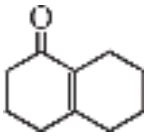
(4) 

8)  , Product (B) is :

(1) 

(2) 

(3) 

(4) 

9) Which reducing agent of the following can be used for the following transformation?
CH_3–CH=CH.COOH → CH_3CH=CH.CH_2OH

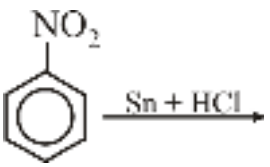
(1) $LiAlH_4$

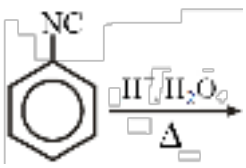
(2) $NaBH_4$

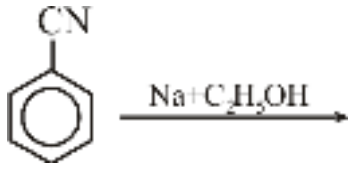
(3) H_2/Ni

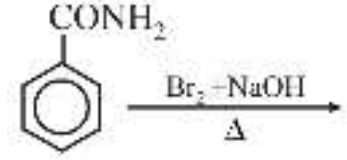
(4) Red P + HI

10) Which of the following reaction will not produce aniline?

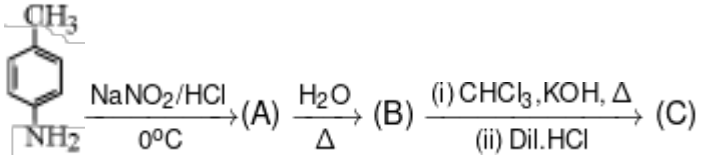
(1) 

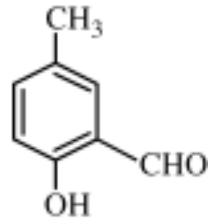
(2) 

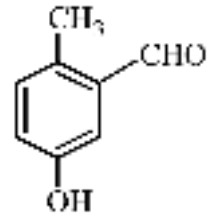
(3) 

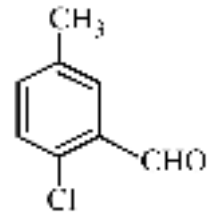
(4) 

11) Identify C in the reactions :

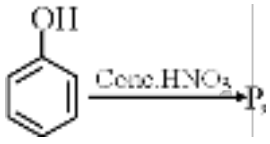


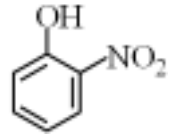
(1) 

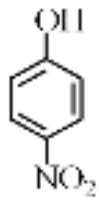
(2) 

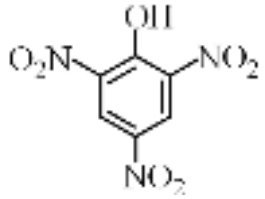
(3) 

(4) None of these

12)  Product 'P' is :

(1) 

(2) 

(3) 

(4) Both 1 and 2

13) Match the vitamins given in Column-I with the deficiency diseases listed in Column-II

Column-I		Column-II	
(a)	Vitamin A	(p)	Osteomalacia
(b)	Vitamin C	(q)	Beri beri
(c)	Vitamin D	(r)	Scurvy
(d)	Vitamin B_1	(s)	Xerophthalmia

(1) a → p; b → q; c → s; d → r

(2) a → s; b → r; c → p; d → q

(3) a → s; b → p; c → r; d → q

(4) a → p; b → r; c → q; d → s

14) Incorrect statement from the following is:-

(1) The non-reducing half in lactose is β-galactose

(2) Cellulose is polymer of β-glucose

(3) Starch is polymer of α-glucose

(4) In amylose glucose residues are connected by 1,6 glycosidic linkage

15) Which of the following pairs give positive Tollen’s test ?

(1) Glucose, Sucrose

(2) Glucose, Fructose

(3) Hexanal, acetophenone

(4) Fructose, sucrose

16)

Which of the following compounds does not show Lassaigne test for nitrogen?

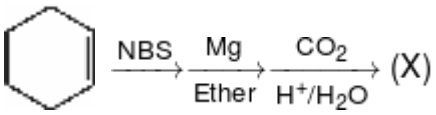
(1) Urea

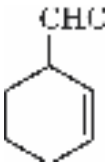
(2) Hydrazine

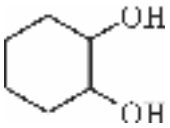
(3) Phenyl hydrazene

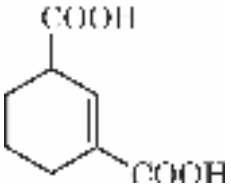
(4) Azo benzene

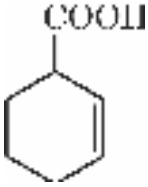
17) In the given reaction, final product (X) will be :

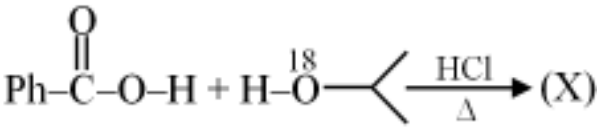


(1) 

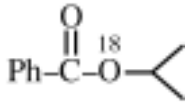
(2) 

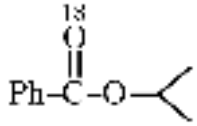
(3) 

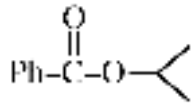
(4) 

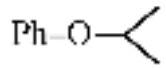
18) 

Major product (X) is :-

(1) 

(2) 

(3) 

(4) 

19) Nitrobenzene can be converted into N-phenyl hydroxyl amine by :-

(1) $LiAlH_4$

(2) $NaBH_4$

(3) Fe / steam

(4) Zn / NH_4Cl

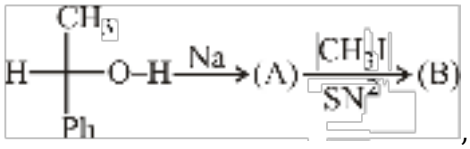
20) **Assertion** :- Aldehyde is more reactive than ketone towards nucleophilic attack.
Reason :- There is less +I effect and less steric hindrance in aldehyde as compared to ketone.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

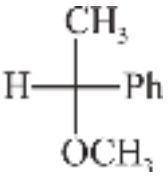
(3) **Assertion** is false but **Reason** is true.

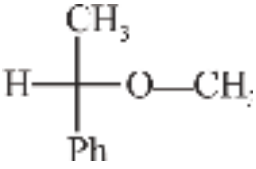
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

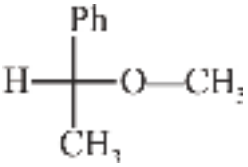
21) 

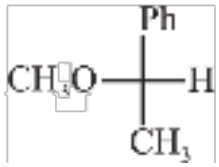
,

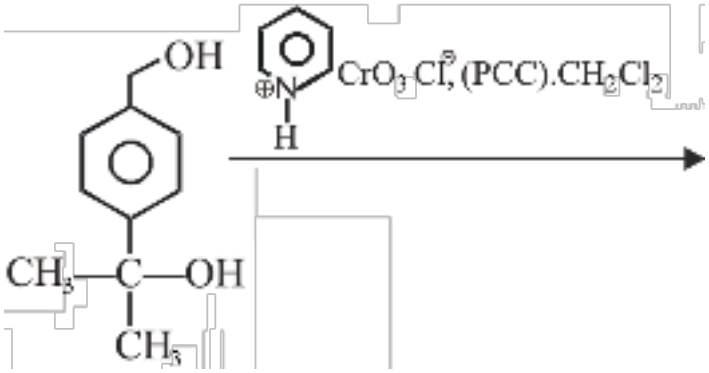
Product (B) is :-

(1) 

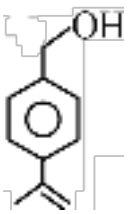
(2) 

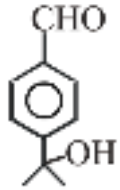
(3) 

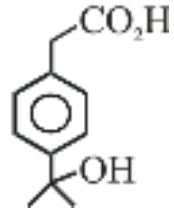
(4) 

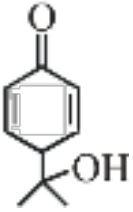
22) 

Product of the reaction is :

(1) 

(2) 

(3) 

(4) 

23) In the reaction

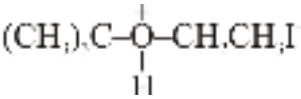
$$(CH_3)_3C-O-CH_2CH_3 + \underset{(1\ \text{mole})}{HI} \xrightarrow{\text{heat}}$$

the product(s) formed is (are)

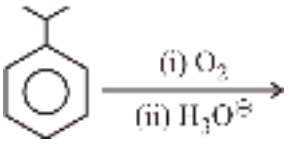
(1) $(CH_3)_3C$–OH and CH_3CH_2I

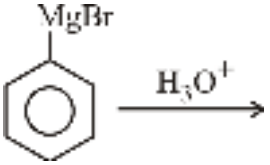
(2) $(CH_3)_3C$–I and CH_3CH_2OH

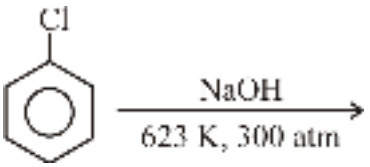
(3) $(CH_3)_3C$–I and CH_3CH_2I

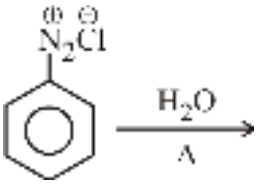
(4) 

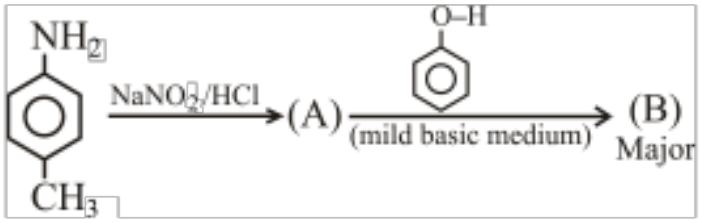
24) In which of the following reaction, phenol is not obtained :-

(1) 

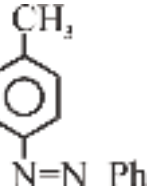
(2) 

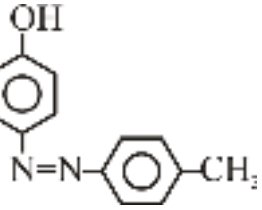
(3) 

(4) 

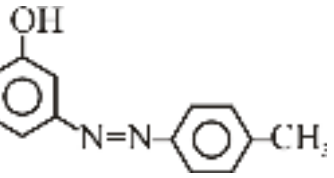
25) 

Product (B) of this reaction is

(1) 

(2) 

(3) 

(4) 

26) The reaction of $RCONH_2$ with a mixture of Br_2 and aqueous KOH gives RNH_2 as the main product. The intermediate not involved this reaction is (are) :-

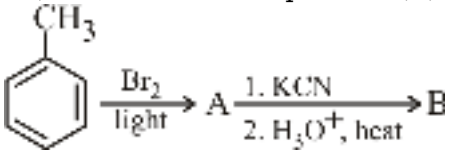
(1) RCONHBr

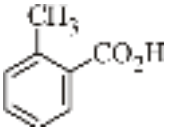
(2) $RCON^-BrK^+$

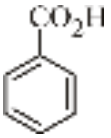
(3) R – N = C = O

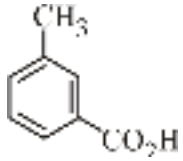
(4) $RCONBr_2$

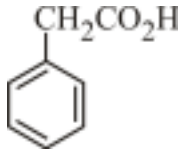
27) What is the final product (B) of this sequence?



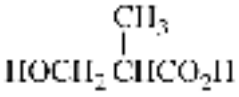
(1) 

(2) 

(3) 

(4) 

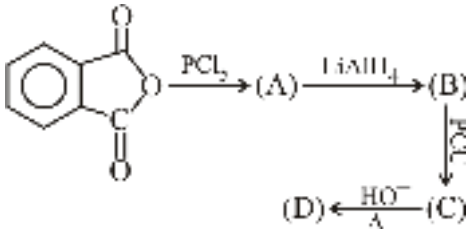
28) A compound (X), $C_4H_8O_3$, liberates CO_2 on reaction with $NaHCO_3$. When (X) is treated with Pyridinium chlorochromate, it is converted into a new compound (Y), $C_4H_6O_3$, which on heating expels CO_2 to form acetone. The compound (X) is:-

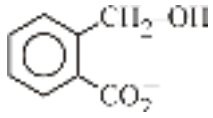
(1) 

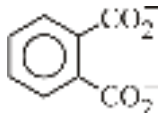
(2) 

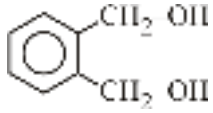
(3) $CH_3CHOHCH_2CO_2H$

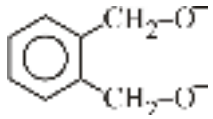
(4) $CH_3CH_2CH(OH)CO_2H$

29) 

(1) 

(2) 

(3) 

(4) 

30)

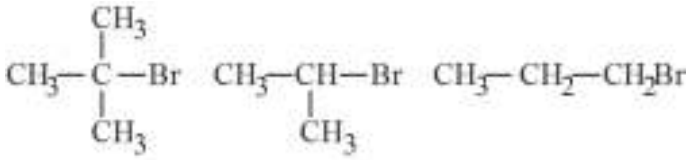
Arrange the acid derivative in the order of nucleophilic substitution reaction :-
(i) Acid halide (ii) Ester
(iii) Anhydride (iv) Amide

(1) i > ii > iii > iv

(2) ii > iii > i > iv

(3) i > iii > ii > iv

(4) iv > ii > iii > i



(a) (b) (c)

31) Reactivity towards S_N^1 :-

(1) a > b > c

(2) b > a > c

(3) b > c > a

(4) c > b > a

32) Amylopectin is composed of

(1) α -D-glucose with C_1–C_4 and C_1–C_6 linkage

(2) α - D-glucose with C_1–C_4 and C_2–C_6 linkage

(3) β - D-glucose with C_1–C_4 and C_2–C_6 linkage

(4) β - D-glucose with C_1– C_4 and C_1–C_6 linkage

33) Acetaldehyde and Benzaldehyde can be differentiated by -

(1) Fehling test

(2) Iodoform test

(3) Tollen's reagent

(4) Both (1) & (2)

34) ***Assertion*** :- Direct nitration of aniline by using conc. HNO_3, meta product is also form as product.
Reason :- In aniline, $–NH_2$ group is meta directing.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

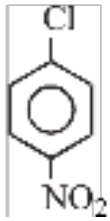
(2) **Assertion** is true but **Reason** is false.

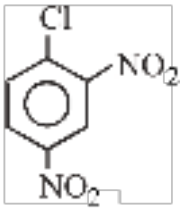
(3) **Assertion** is false but **Reason** is true.

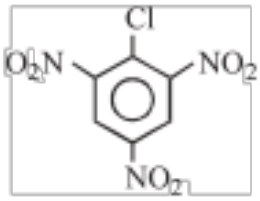
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

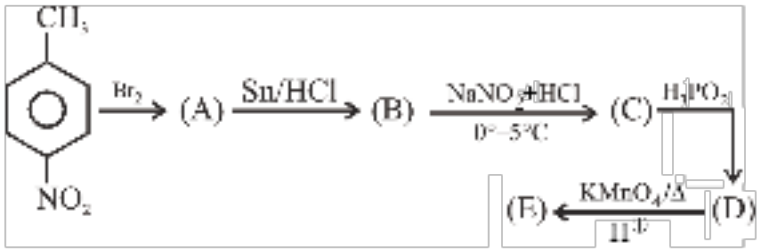
35) Which of the following is most easily hydrolysed :-

(1) 

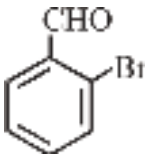
(2) 

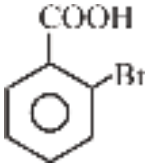
(3) 

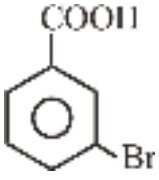
(4) 

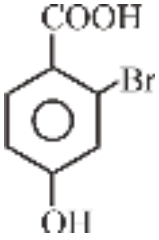
36) 

End product 'E' will be :-

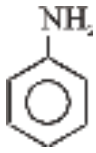
(1) 

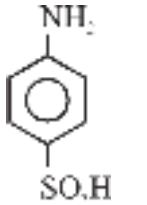
(2) 

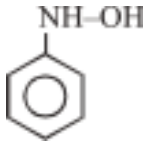
(3) 

(4) 

37) Benzene $\xrightarrow{HNO_3+H_2SO_4}$ A $\xrightarrow[\text{electrolysis}]{\text{dil } H_2SO_4}$ B; B is :-

(1) 

(2) 

(3) 

(4) 

38) ***Assertion*** :- Aniline does not give Friedel Craft reaction.
Reason :- Reagent used ($AlCl_3$) in Friedel Craft reaction is a lewis acid and attacks on the lone pair of Aniline thus converting it into deactivating group.

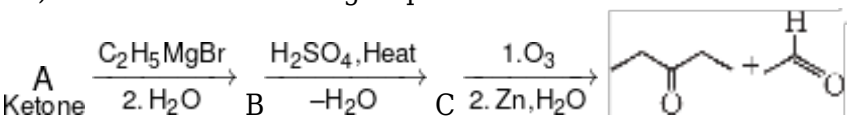
(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

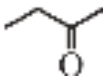
(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

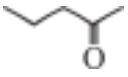
39) Consider the following sequence of reactions



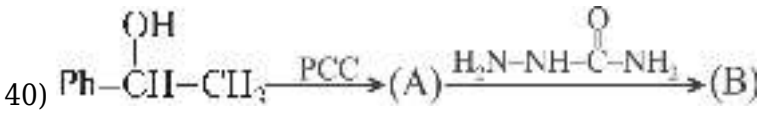
The ketone (A) is :

(1) 

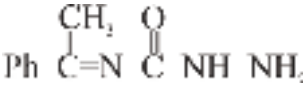
(2) 

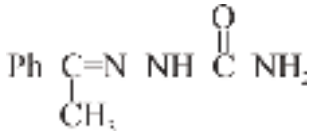
(3) 

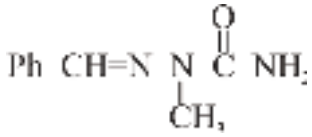
(4) 

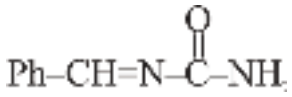
40) 

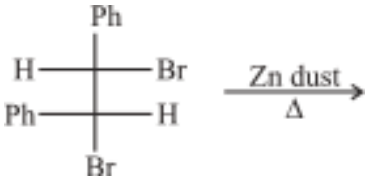
Product (B) is :-

(1) 

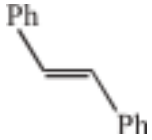
(2) 

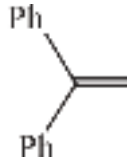
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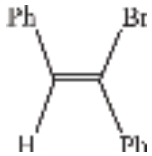
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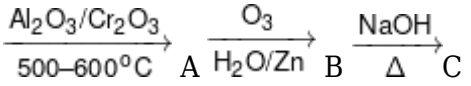
41)  Major product :-

(1) 

(2) 

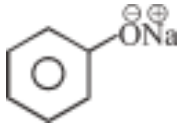
(3) 

(4) 

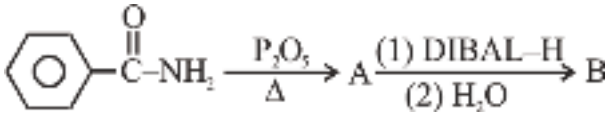
42) n-Hexane  product (C) will be :-

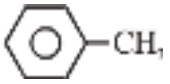
(1) 

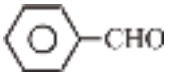
(2) 

(3) 

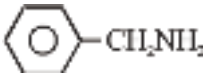
(4) 

43) 
Identify product B in above reaction ?

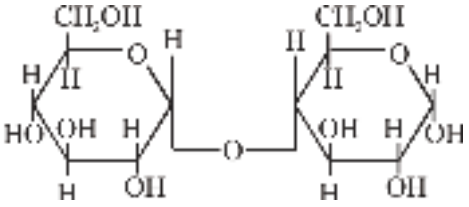
(1) 

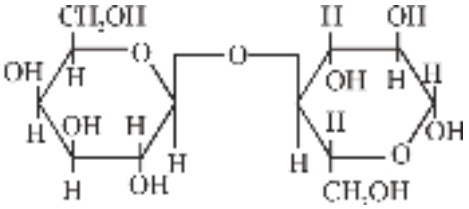
(2) 

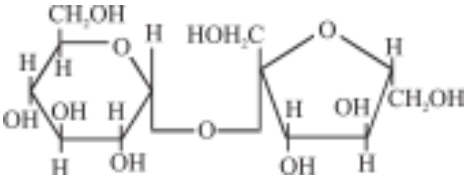
(3) 

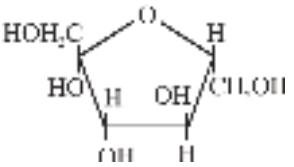
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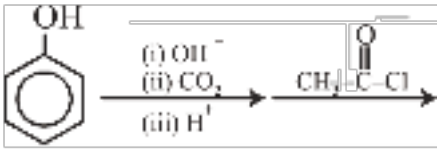
44) Which of the following comopound will not give test with tollen's reagent :-

(1) 

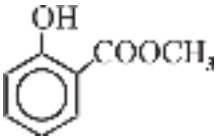
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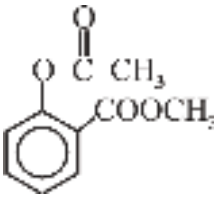
(3) 

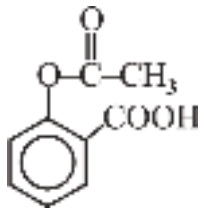
(4) 

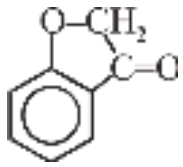
45)  (A) drug.

Drug A is :-

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) **Assertion (A) :** All animals depend on plants (directly or indirectly) for their food needs, they are hence called consumers and also primary consumers if they feed on plants and if the animals eat other animals which in turn eat the plants (or their produce) they are called secondary consumers.
Reason (R) : Pyramid of energy is always upright can never be inverted, because when energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step.
In the light of above statements choose the correct answer from the options given below.

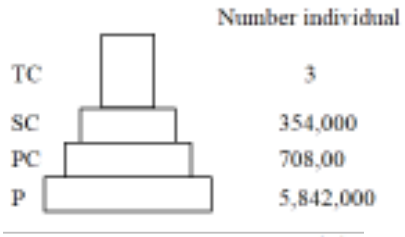
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

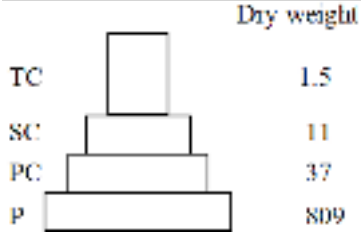
(2) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**.

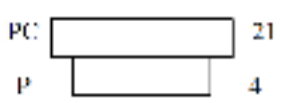
(3) **(A)** is true but **(R)** is false

(4) **(A)** is false but **(R)** is true

2) Which of the following pyramid represent small standing crop (Biomass) of phytoplankton supports large standing crop of zooplankton ?

(1) 

(2) 

(3) 

(4) 

3)

Match List-I with List-II :-

	List-I		List-II
(A)	Primary production	(I)	Available biomass for heterotrophs
(B)	Net primary productivity	(II)	Rate of formation of new organic matter by consumer
(C)	Secondary productivity	(III)	R
(D)	Respiration losses	(IV)	Organic matter produced by plant per unit area over a time

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-III, C-I, D-IV

(3) A-IV, B-I, C-III, D-II

(4) A-IV, B-I, C-II, D-III

4) Choose the incorrect statement :-

(1) All animals depend on plant directly or indirectly for their food needs.

(2) Primary consumers can be herbivores.

(3) The consumer that feed on these herbivores are carnivores, or more correctly primary carnivores

(4) These animals that depend on the herbivores for food are called as secondary carnivores.

5) Match List-I with List-II

	List-I		List-II
(A)	Fragmentation	(I)	Breakdown of detritus in to smaller particles.
(B)	Catabolism	(II)	Release of inorganic nutrients in soil
(C)	Mineralisation	(III)	Degrade detritus in simpler inorganic substances.

Choose the correct answer from the options given below :-

(1) A-I, B-III, C-II

(2) A-I, B-II, C-III

(3) A-II, B-III, C-I

(4) A-III, B-I, C-II

6) The amount of energy decrease at successive trophic levels due to :-

(1) 80% energy loss at each trophic level
(2) 85% energy loss at each trophic level
(3) 75% energy loss at each trophic level
(4) 90% energy loss at each trophic level

7) Of the total incident solar radiation (ISR) Photosynthetic active radiation (PAR) is :-

(1) Less than 50%
(2) More than 50%
(3) 2-10%
(4) 1-5%

8) What shape of pyramid of number would you obtain if large number of insects feeding on a single tree?

(1) Upright
(2) Inverted
(3) Spindle
(4) May be upright or spindle

9) Read statements carefully :
Statement-I : Identification and enumeration of plant and animal species of an ecosystem gives its species composition.
Statement-II : The chain or web is formed because of this food interdependency in ecosystem.

(1) **Statements I & II** both are correct
(2) **Statements I & II** both are incorrect
(3) **Statement-I** is correct only
(4) **Statement-II** is incorrect only

10) What is the main function of a detritivore in an ecosystem :

(1) Pollination
(2) Parasitism
(3) Pathogenic
(4) Nutrient recycling

11) Which of the following is a limitation of ecological pyramids ?

(1) Saprophytes are given a place in ecological pyramids.
(2) It assumes a simple food chain, that almost never exist in nature.
(3) It accomodates a food web.
(4) It takes into account the same species belonging to two or more trophic levels.

12) Read the column-I, II, III and select the correct one :-

	Column-I		Column-II		Column-III
a.	Scavengers	(i)	Autotroph	P	Phytoplankton
b.	Producers	(ii)	Feed on dead body	Q	Vultures
c.	National parks	(iii)	Three in India	R	Indoburma
d.	Hot spot	(iv)	90 in India	S	Insitu conservation

(1) a-(ii)-Q, b-(i)-P, c-(iv)-S, d-(iii)-R

(2) a-(i)-P, b-(ii)-Q, c-(iv)-S, d-(iii)-R

(3) a-(ii)-Q, b-(i)-P, c-(iii)-S, d-(iv)-R

(4) a-(i)-Q, b-(ii)-P, c-(iii)-S, d-(iv)-R

13) Which of the following is not the part of evil quartet :-
(A) Habitat loss and fragmentation
(B) Over exploitation
(C) Alien species invasion
(D) Pollution
Choose the correct answer from the options given below :-

(1) A, B and C

(2) B and C only

(3) D only

(4) C and D only

14) Although all the biodiversity hot spots put together cover less than……A……..percent of earth's land area, the no. of species they collectively harbour is extremely high and strict protection of these hotspots could reduce the ongoing mass extinction by almost…..B….percent.
A & B are respectively :-

(1) A 2%, B 30%

(2) A 1%, B 30%

(3) A 2%, B 20%

(4) A 1%, B 20%

15) When was the historic convention on biological diversity earth summit held in Rio de janeiro :-

(1) 1992

(2) 2002

(3) 1982

(4) 2012

16) The interaction confers benefits on both the interacting species ?

(1) Protocooperation

(2) Mutualism

(3) (1) & (2) both

(4) Competition

17) How many among the given statements are correct.
(i) Commensalism is also known as facultative mutualism.
(ii) Commensalism confers benefits on both the interacting species and they should live together for their survival.
(iii) Epiphyte on mango branch is example of commensalism.
(iv) Mutualism shows co-evolution.

(1) iii, iv
(2) i, iv
(3) ii, iv
(4) i, ii

18) Match the following correctly.

i	Commensalism	a	Ammensalism
ii	Mutualism	b	Cattle egret bird and grazing cattle
iii	Predation	c	Wasp-fig and Orchid-bumble bee
iv	–/0	d	Sparrow eating seeds

(1) i – b, ii – c, iii – d, iv – a
(2) i – a, ii – b, iii – c, iv – d
(3) i – d, ii – c, iii – b, iv – a
(4) i – d, ii – b, iii – c, iv – a

19) Select the correct option with respect to parasitism ?

(1) Copepods on marine fishes.
(2) Sea anemone and Clown fish
(3) Abingdon tortoise and goats
(4) Fig tree and Wasps

20) Vertical distribution of different species occupying different level is called _____.

(1) Succession
(2) Stratification
(3) Physiognomy
(4) Edge Effect

21) **Statement–I** :- Species diversity depends on size of the area, type of soil and climate.
Statement–II :- $\log S = \log C + Z \log A$
Choose the correct option :-

(1) Both **Statement I** and **Statement II** are incorrect.
(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

22) Read the following statement and select correct ones :-
(a) The most spectacular and evolutionary fascinating examples of mutualism are found in plant and animal relationship.
(b) The tiger census in our national parks and tiger reserves is often based on pug marks and fecal pellets.
(c) Bacteria in culture plate is not a example of population.
(d) The size of population for any species is a static parameter.

(1) (a) & (b)

(2) (b) & (c)

(3) (c) & (d)

(4) (a) & (d)

23) Match the column-I, II and III and select the correct one :-

	Column-I		**Column-II**		**Column-III**
A	Commensalism	(i)	One harmed other unaffected	P.	Orchid on mango
B	Parasitism	(ii)	One benefitted other unaffected	Q.	Cuckoo and Crow
C	Mutualism	(iii)	Both are benefitted	R.	Mycorrhiza
D	Amensalism	(iv)	One benefitted other harmed	S.	*Parthenium*

(1) A-(ii)-P, B-(iv)-Q, C-(iii)-R, D-(i)-S

(2) A-(ii)-P, B-(iv)-Q, C-(i)-S, D-(i)-R

(3) A-(iii)-R, B-(ii)-P, C-(iv)-Q, D-(i)-S

(4) A-(iv)-P, B-(iii)-Q, C-(i)-R, D-(ii)-S

24) Given below are two statements :
Statement-I : When we think of predator and prey, most probably it is the tiger and deer that readily come to our mind.
Statement-II : A sparrow eating any seed is no less a predator.
In the light of above statement, choose the correct answer from the option given below :

(1) Both **Statement I** and **Statement II** are false

(2) **Statement II** is false but **Statement I** is true

(3) Both **Statement I** and **Statement II** are true

(4) **Statement II** is true and **Statement I** is false

25) Flamingoes and resident fishes compete for their common food. What food was that?

(1) Small size fishes

(2) Phytoplankton

(3) Zooplankton

(4) All of these

26) Given below are two statement :
Statement-I : Lichen represent an intimate mutualistic relationship between a fungus and photosynthesising algae or cyanobacteria.
Statement-II : Plant offer rewards or fees only in the form of nectar.

(1) **Statement I** is false and **Statement II** is true

(2) **Statement I** is true and **Statement II** is false

(3) Both **Statement I** and **Statement II** are true

(4) Both **Statement I** and **Statement II** are false

27) Given below are two statements : One is labelled as assertion (A) and the other is labelled as reason (R).
Assertion (A) : Majority of the parasites harm the host, they may reduce the survival, growth and reproduction of the host and reduce the population density.
Reason (R) : Liverfluke is a ectoparasite.
In the light of the above statements, choose the correct answer from the option given below :-

(1) Both A and R are true but R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of R.

(3) A is true but R is false.

(4) A is false but R is true.

28) Two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually. This is called -

(1) Competitive exclusion principle

(2) Competitive release

(3) Resource partitioning

(4) Camouflage

29) Kangaroo in Australia is an example of _____.

(1) Endemic species

(2) Keystone species

(3) critical link species

(4) Extinct species.

30) **Assertion :** When resources are unlimited, the population grows in an exponential fashion.
Reason : When resources are unlimited, each species has the ability to realise fully its innate potential to grow in numbers.
Select the correct answer from the options given below :-

(1) Both assertion and reason are true and reason is correct explanation of assertion.

(2) Assertion is true both reason is false.

(3) Both assertion and reason are false.

(4) Both assertion and reason are true but reason is not correct explanation of assertion.

31) If in a pond there were 20 lotus plants last year and through reproduction 8 new plants are added, calculate the birth rate per year :-

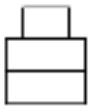
(1) 0.4

(2) 0.5

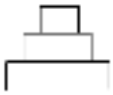
(3) 0.6

(4) 0.8

32) Which of the following age pyramids suits fit for Indian population

(1) 

(2) 

(3) 

(4) None of these

33) Which of the following processes results into the increase in population density.

(1) Natality and Mortality

(2) Natality and Immigration

(3) Natality and Emigration

(4) Mortality and Immigration

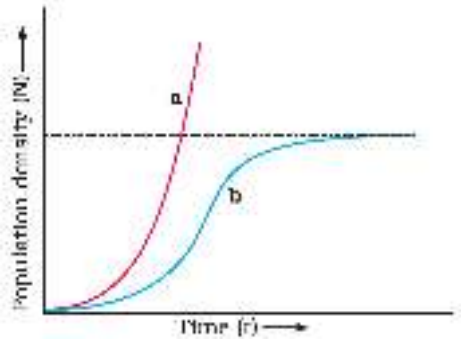
34) Read statement carefully :-
a. Species evolve to maximise their reproductive fitness in the habitat in which they live.
b. K-selected species typically exhibit-slow population growth.
c. r-selected species produce large number of progeny
d. Pacific salmon fish breed many time during their lifetime.
How many statement is/are incorrect :

(1) One

(2) Two

(3) Three

(4) All are correct

35) 

Identify the correct statement regarding the figure given above :-
(A) Curve 'a' shows logistic growth.
(B) Curve 'b' is represented by $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$
(C) Curve 'b' is obtained when responses are limiting the growth.
(D) Curve 'a' is more realistic in nature and is shown by maximum species.

(1) A and C

(2) B and C

(3) A and B

(4) C and D

36) Identify the correct statements :-
(A) A population growing in a habitat with limited resources show initially log phase, followed by phase of deacceleration and acceleration and finally an asymptote.
(B) When resources in the habitat are unlimited, each species has the ability to realise fully its innate potential to grow in number, then population grow in geometric fashion.
(C) Under normal conditions, births and deaths are the most important factors influencing population density.
(D) The size of a population for any species is a static parameters.
(E) An individual may have birth rate and death rate, but a population has birth and death.
Choose the correct answer from the option given below :-

(1) A, D, C only

(2) A, C, E only

(3) B, C only

(4) C, E only

37) Which of the following option represent omnivores?

(1) Insects, birds, mammals and Molluscs

(2) Birds, fishes and wolf

(3) Cockroaches, crows, peacock

(4) Zooplankton, Sparrow, Human, Molluscs

38) **Assertion (A) :** In many cultures, tracts of forest were set aside, and all the trees and wildlife within were worshipped and given total protection.
Reason (R) : Sacred groves are examples of in-situ conservation.

(1) Both A and R are true and R is correct explanation of A.

(2) Both A and R are true but R is not correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

39) India, with much of it's land area in the tropical latitudes has _____ species of birds.
Complete the statement using most appropriate option among the following options.

(1) less than 400

(2) more than 1400

(3) more than 1200

(4) only 3000

40) Scientists estimate that in amazon rain forests there might be at least _____waiting to be discovered and named.
Complete the above statement by using most appropriate option among the following options.

(1) Four million insect species

(2) Two million insect species

(3) Two billion Angiosperms

(4) Four billion bird species

41) Choose the incorrect statement

(1) Communities with more species, generally tend to be more stable than those with less species.

(2) A stable community show too much variation in productivity from year to year.

(3) A stable community must be either resistant or resilient to occasional disturbances (natural or man-made)

(4) A stable community must also be resistant to invasions by alien species.

42) Match **column–I** with **column–II** with respect to rivet popper hypothesis and choose the most appropriate match.

	Column–I		**Column–II**
(i)	Airplane	(a)	Key-stone species
(ii)	Rivets	(b)	Ecosystem
(iii)	Rivets on wings	(c)	Species

(1) (i)–(a); (ii)–(b), (iii)–(c)

(2) (i)–(c); (ii)–(b), (iii)–(a)

(3) (i)–(b); (ii)–(c), (iii)–(a)

(4) (i)–(b); (ii)–(a), (iii)–(c)

43) _____Showed that in his experiments, increased diversity contributed to higher productivity.

(1) Paul Ehrlich

(2) Robert May

(3) David Tilman

(4) Edward Wilson

44) Select incorrect statement regarding "biodiversity hotspots".

(1) Very high levels of species richness

(2) High degree of endemism

(3) Total number in the world -25

(4) Cover less than 2 percent of earth's land area.

45) How many of the following are methods of *in-situ* conservation?
Biodiversity hot spots, zoological parks, National parks, wildlife sanctuaries, Biosphere reserves, Botanical gardens, sacred groves, wildlife safari park.

(1) Four

(2) Five

(3) Six

(4) Seven

46) Mutations are ____(A)____ and ____(B)_____ while Darwinian variations are _____(C)_____ and _____(D)_____. Fill in the blanks for A, B, C and D from the options given in the table below:

	A	B	C	D
(1)	Small	Directional	Random	Directionless
(2)	Directional	Large	Small	Directionless
(3)	Random	Directionless	Small	Directional
(4)	Large	Directional	Small	Directional

(1) 1

(2) 2

(3) 3

(4) 4

47) Read the following statements and choose the correct option.
(A) Evolution by natural selection would have started when multicellular forms of life with differences in metabolic capability originated on earth.
(B) Fitness is the end result of the ability to adapt and get selected by nature.
(C) Work of Thomas Malthus on populations influenced Lamarck.
(D) Theoretically, population size will grow exponentially if everybody reproduced maximally.

	(A)	(B)	(C)	(D)
(1)	T	T	F	F
(2)	F	T	T	F
(3)	T	F	T	T

(4)	F	T	F	T

(1) 1

(2) 2

(3) 3

(4) 4

48) Some examples are given below:
(a) Forelimbs of mammals
(b) Thorn of Bougainvillea and tendril of cucurbita
(c) Wings of birds and butterfly
(d) Eyes of the octopus and mammals
(e) Flippers of penguins and dolphins
(f) Sweet potato and potato
How many examples among these are the examples of analogous structures in animals?
Options :

(1) c, d, e, f

(2) a, b, c

(3) c, d, e

(4) e, f

49) Match the following :

(A)	Louis Pasteur	(I)	First life came from non-living matter
(B)	Oparin and Haldane	(II)	Synthesis of amino acids
(C)	Miller and Uray	(III)	Darwin like thoughts
(D)	Alfred Wallace	(IV)	Life did not came from killed yeast
(E)	Darwin	(V)	Wallace like thoughts
(F)	Lamark	(VI)	Inheritance of acquired characters

(1) A - IV, B - II, C - I, D - III, E - VI, F - V

(2) A - IV, B - I, C - II, D - III, E - V, F - VI

(3) A - VI, B - III, C - I, D - II, E - IV, F - V

(4) A - I, B - II, C - III, D - IV, E - VI, F - V

50) **Assertion :** Lichens can be used as pollution indicators
Reason : Lichens will not grow in areas that are pollution free

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false statements

51) Match List-I with List-II :-

	List-I		List-II
(A)	Gene frequency	(I)	When gene migration happen multiple time
(B)	Gene flow	(II)	Relative proportion of a specific allele within the total gene pool of a population
(C)	Gene pool	(III)	When section of population gets isolated from original population
(D)	Founder effect	(IV)	Total number of gene and their alleles

Choose the correct answer from the option given below :-

(1) A-I, B-II, C-IV, D-III

(2) A-II, B-I, C-IV, D-III

(3) A-I, B-II, C-III, D-IV

(4) A-IV, B-III, C-II, D-I

52) Which of the following statement is incorrect?

(1) Big Bang theory talks of a singular huge explosion unimaginable in physical terms.

(2) In the solar system of the milky way galaxy, earth was supposed to have been formed about 4.5 billion years back.

(3) Water vapour, methane, carbon dioxide and ammonia released from molten mass covered the primitive earth surface.

(4) Life appeared 4 billion years after the formation of earth.

53) Read the following statement :

Statement-I : According to theory of spontaneous generation, life came out of decaying and rotting matter like straw, mud, etc.

Statement-II : Louis Pasteur by careful experimentation demonstrated that life comes only from pre-existing life.

In the light of the above statements, choose the correct answer from the options given below :

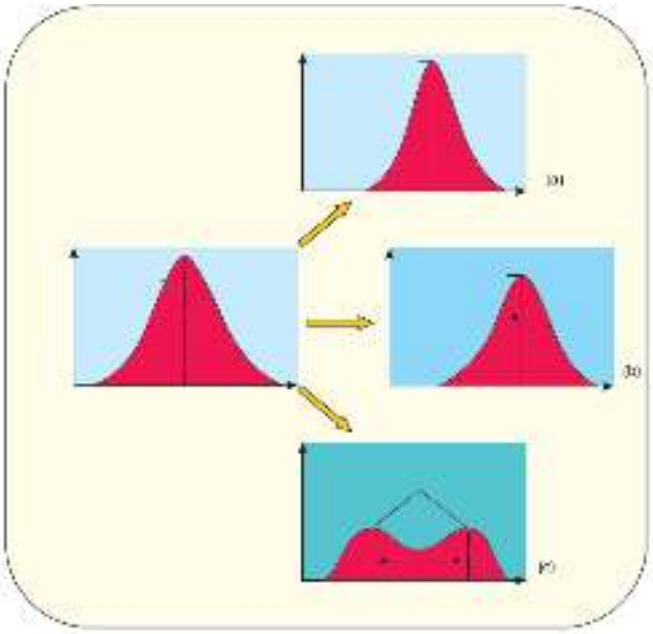
(1) Both Statements I and Statement II are correct.

(2) Both Statements I and Statement II are incorrect.

(3) Statement I is correct and Statement II is incorrect.

(4) Statement II is correct and Statement I is incorrect

54) Recognise the figure and find out the correct matching :-



(1) b–directional, a–disruptive, c–stabilising

(2) b–directional, c–disruptive, a–stabilising

(3) c–directional, b–disruptive, a–stabilising

(4) a–directional, c–disruptive, b–stabilising

55) In a large population, 16% individuals are homozygous dominant, 42% individuals are heterozygous and remaining are homozygous recessive. Calculate the frequency of dominant and recessive alleles.

(1) Dominant allele - 37%, Recessive allele - 63%

(2) Dominant allele - 63%, Recessive allele - 37%

(3) Dominant allele - 40%, Recessive allele - 60%

(4) Dominant allele - 60%, Recessive allele - 40%

56) Prehistoric cave art develop about :

(1) 10,000 ya

(2) 18000 ya

(3) 15000 ya

(4) 34000 ya

57) In 1938 a fish caught in South Africa happened to be a Coelacanth which was thought to be extinct. These animals called lobefins evolved into the first :-

(1) Amphibians

(2) Jawless fish

(3) Reptiles

(4) Birds

58) Which of the following method is not used to determine the age of the fossils?

(1) Electron spin resonance

(2) Autoradiography

(3) Carbon dating method

(4) Uranium lead method

59) Stanley Miller used following gases in closed container :-

(1) CH_4, H_2, NH_3 & vapour at 800°C

(2) CH_4, H_2, HCN & vapour at 800°C

(3) CH_4, H_2, NH_3 & vapuor at 100°C

(4) CH_4, NH_3, HCN & vapour at 500°

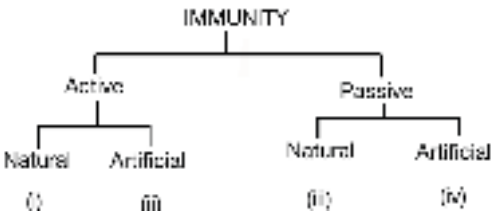
60) Due to continental drift which mammal survived only in Australia because of lack of competition from any other mammal :

(1) Horse

(2) Pouched mammals

(3) Hippopotanmus

(4) `Penguins

61) 

(1) i-Infection, ii-Antiserum, iii-IgA, iv-Vaccine

(2) i-IgA, ii-Vaccine, iii-Infection, iv-Antiserum

(3) i-Infection, ii-Vaccine, iii-IgA, iv-Antiserum

(4) i-IgG, ii-Vaccine, iii-Infection, iv-Antiserum

62) How many of the given disease are avoid by control and eliminate the vectar and their breeding places.
Dengue, Filariasis, Malaria, Ringworm, Chikungunya, Ascariasis, Amoebiasis

(1) four

(2) Three

(3) five

(4) Six

63) Physiological barriers of innate immunity include
(a) Macrophages
(b) NK cells
(c) Mucus coating of the epithelium lining the respiratory tract
(d) Saliva in mouth
(e) Acid in stomach

Choose the correct answer from the options given below.

(1) (a) and (b) only

(2) (a), (b), (c) and (e)

(3) (c), (d) and (e)

(4) (d) and (e) only

64) Which of the following statements regarding drugs and alcohol abuse are true?
(a) Heroin is an opioid that depresses the central nervous system.
(b) Cocaine is a stimulant that affects the central nervous system.
(c) Long-term alcohol abuse can lead to liver cirrhosis.
(d) Smoking tobacco is linked to an increased risk of lung cancer.
(e) Barbiturates are used to treat bacterial infections.
Choose the **correct** answer from the options given below :

(1) a, c, e only

(2) a, b, c, d only

(3) a, b, e only

(4) b, c, d, e only

65) Read the following statements and choose the True (T) and False (F) about them from given options-
(A) Virus infected cells secrete proteins called antigen
(B) Each antibody molecule has two longer light chains and two small heavy chains.
(C) The principle of vaccination is based on the property of memory of the immune system.
(D) Both innate and acquired immunity are pathogen specific.

	(A)	(B)	(C)	(D)
1	T	T	T	F
2	F	T	T	F
3	F	F	T	F
4	F	T	F	T

(1) 1

(2) 2

(3) 3

(4) 4

66) In the given list how many are included in Cannabinoids?
Marijuana, Morphine, Hashish, Charas, Ganja

(1) 4

(2) 3

(3) 2

(4) 1

67) A patient consumed contaminated food few days before, now he is suffering from sustained high fever, headache, weakness and loss of appetite. What can be the probable disease?

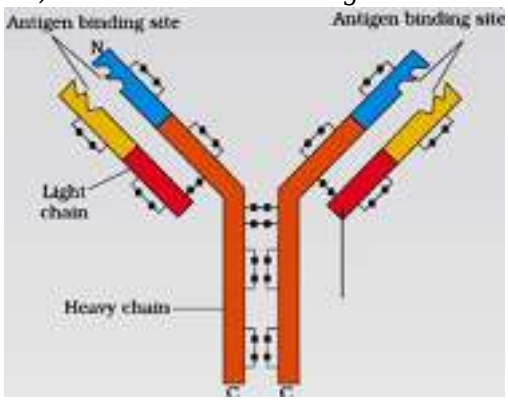
(1) Typhoid

(2) Pneumonia

(3) Diphtheria

(4) Tuberculosis

68) Which of the following statement is false for the structure given below :



(1) These are secreted by T-cells

(2) They have four peptide chains

(3) Represented by H_2L_2

(4) These are found in blood so response is termed as humoral immune response

69) Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : Houseflies act as mechanical carriers in amoebiasis.
Reason R : Houseflies can transmit the parasite from feces of infected persons to food and food products.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) A is correct but R is not correct

(2) A is not correct but R is correct

(3) Both A and R are correct and R is the correct explanation of A

(4) Both A and R are correct but R is NOT the correct explanation of A.

70) **Assertion :** In AIDS, helper T-lymphocytes decreases.
Reason : HIV enters into helper T-lymphocytes, replicates and produces progeny viruses and destroyed them.

(1) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is correct but **Reason** is incorrect.

(4) **Assertion** is incorrect but **Reason** is correct.

71) **Assertion :** Colostrum provides passive immunity.
Reason : Colostrum is yellowish fluid, secreted by mother during initial days of location has abundant antibodies (IgA) to protect the infant.

(1) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is correct but **Reason** is incorrect.
(4) **Assertion** is incorrect but **Reason** is correct.

72) Given below are two statements :
Statement-I : The exaggerated response of the immune system to certain antigens present in the environment is called auto-immunity.
Statement-II : The antibodies produced to allergens are of IgA type.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement-I is true but Statement-II is false
(2) Statement-I is false but Statement-II is true
(3) Both Statement-I and Statement-II are true
(4) Both Statement-I and Statement-II are false

73) Given below are two statements-
Statement-I : Certain types of leukocytes of our body like polymorpho-nuclear leukocytes and monocytes and natural killer cells in the blood as well as macrophages in tissues can phagocytose and destroy microbes.
Statement-II : Acid in the stomach, saliva in the mouth and tears from eyes-all prevent microbial growth.
In the light of the above statement choose correct answer from the options given below-

(1) Statement-I and statement-II both are incorrect.
(2) Statement-I is correct but statement-II is incorrect.
(3) Statement-I is incorrect but statement-II is correct.
(4) Statement-I and statement-II both are correct.

74) Given below are two statements:
Statement I: Health simply means 'absence of disease' or 'physical fitness'.
Statement II: Health could be defined as a state of complete physical, mental and social well-being.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Statement I is true but Statement II is false
(2) Statement I is false but Statement II is true
(3) Both Statement I and Statement II are true
(4) Both Statement I and Statement II are false

75) Match the list-I with list-II and choose correct answer about them:

	List -I		List-II
(A)	Cytokine barrier	(i)	Uses strong magnetic field and non-Ionising radiations.
(B)	Graft rejection	(ii)	Thymus and bone marrow
(C)	Primary lymphoid organs	(iii)	Cell mediated immunity
(D)	MRI	(iv)	Interferons

(1) A – i, B – ii, C – iii, D – iv

(2) A – iv, B – iii, C – ii, D – i

(3) A – iii, B – iv, C – i, D – ii

(4) A – ii, B – i, C – iv, D – iii

76)

Match the column-I with column-II

	Column-I		**Column-II**
(A)	Ascariasis	(i)	Bite by female mosquito vectors
(B)	Elephantiasis	(ii)	Contaminated food & water.
(C)	Ringworm	(iii)	Using towel of infected individual
(D)	Amoebiasis	(iv)	Contaminated water with eggs

(1) A-ii, B-i, C-iii, D-iv

(2) A-iv, B-i, C-iii, D-ii

(3) A-ii, B-iii, C-i, D-iv

(4) A-ii, B-iv, C-iii, D-i

77)

Match **List-I** with **List-II.**

	List-I		**List-II**
A.	Malignant tumors	I.	Destroy tumors
B.	MALT	II.	AIDS
C.	NACO	III.	Metastasis
D.	α-Interferons	IV.	Lymphoid tissue

Choose the **correct** answer from the options given below :

(1) A-III, B-IV, C-II, D-I

(2) A-IV, B-III, C-II, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-III, B-I, C-IV, D-II

78) Which measures are essential for control of disease like typhoid, amoebiasis and ascariasis?

(1) Proper disposal of waste and excreta.
(2) Periodic cleaning and disinfection of water reservoirs.
(3) Observing standard practices of hygiene in public catering.
(4) All of the above

79) Which of the following statement is false ?

(1) Because of perceived benefits, drugs are frequently used repeatedly.
(2) Use of drugs even once, can be a fore-runner to addiction.
(3) Excessive dosage of cocaine causes hallucination.
(4) Cocaine is obtained from a plant native to south Africa.

80)

Which of the following statement is correct ?

(1) Cancer cells show a property called contact inhibition.
(2) Malignant tumour normally remain confined to their original location
(3) Metastasis is most feared property of benign tumour
(4) Malignant tumors are a mass of proliferating cells called neoplastic cells.

81) Use of nicotine either by smoking or chewing or snuffing is associated with high blood pressure because–

(1) Nicotine directly acts on baroreceptors on the arterial walls
(2) Nicotine stimulates the secretion of adrenaline and nor-adrenaline
(3) It reduces concentration of haembound oxygen
(4) It suppresses the limbic system

82) Which of the following drug is given to patients who have undergone surgery?

(1) Amphetamines
(2) Caffeine
(3) Barbiturates
(4) Morphine

83) Which of the following drugs are abused by sport persons to enchance their performance ?

(1) Narcotic analgesics
(2) Anabolic steroids
(3) Diuretics
(4) All of these

84) Most cancers are treated by combination of :-
(a) Surgery

(b) Radiotherapy
(c) Chemotherapy

(1) a and b
(2) a and c
(3) b and c
(4) a, b and c

85) Opportunistic infections appear in AIDS patient due to low level of -

(1) B-lymphocyte
(2) T-killer cell
(3) Natural killer
(4) T-helper

86) Identify the correct pair of malignant malaria and cause of high fever.

(1) Plasmodium vivax / Haemozoin
(2) Plasmodium falciparum / Haemozoin
(3) Plasmodium malaria / Haemoglobin
(4) Plasmodium malaria / Haemozoin

87) Internal bleeding, muscular pain, fever, anaemia are symptoms of :

(1) Amoebiasis
(2) Filariasis
(3) Ascariasis
(4) Dysentry

88) In severe cases of which disease finger nails may turn gray to bluish in colour.

(1) Typhoid
(2) Malaria
(3) Pneumonia
(4) Common cold

89) The use of vaccines and immunisation programmes have enabled us to completely eradicate a deadly disease like

(1) Diphtheria
(2) Rabies
(3) Pneumonia
(4) Smallpox

90) What is not true for acquired immunity ?

(1) It can recognize a vast variety of micro-organisms
(2) It has immunological memory against antigens
(3) It is the first line of defense against pathogens
(4) It can recognize self and non-self cells

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	4	1	4	4	1	1	4	3	4	2	2	1	2	4	3	3	1	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	3	2	3	3	3	2	1	1	1	1	4	1	3	3	3	2	2	4
Q.	41	42	43	44	45															
A.	3	2	1	1	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	3	3	1	1	4	3	2	1	3	1	3	2	4	2	2	4	1	4	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	2	2	2	2	4	4	3	1	3	1	1	4	2	4	2	1	4	2	2
Q.	86	87	88	89	90															
A.	2	2	2	3	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	4	4	1	4	1	2	1	4	2	1	3	1	1	3	1	1	1	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	1	1	3	3	2	3	1	1	1	1	3	2	1	2	3	3	2	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	3	3	2	3	4	1	2	3	2	4	1	2	1	2	1	2	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	1	4	2	3	1	1	1	3	1	1	4	4	2	2	2	1	4	4	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	4	4	4	4	2	3	3	4	3										

SOLUTIONS

PHYSICS

1)

Condition for bright fringe of λ_1 and λ_2 to coincide, $n_1\lambda_1 = n_2\lambda_2$

$$\frac{n_1}{n_2} = \frac{\lambda_2}{\lambda_1} = \frac{7000\,A^\circ}{6000\,A^\circ}$$

$n_1 = 7$, $n_2 = 6$

2) $\mu = \tan i_p = \sqrt{3}$

$\theta_C = \sin^{-1}\left(\frac{1}{\mu}\right) \Rightarrow \theta_C = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

3)

$A_1 = a_0 \quad A_2 = 2a_0$

$I_1 = I_0 \quad I_2 = 4I_0$

$I_R = I_0 + 4I_0 + 2\sqrt{I_0}\sqrt{4I_0}\cos\phi$

for I_{max}, $\cos\phi = 1$

$I_m = 9I^0$ or $I^0 = \frac{I_m}{9}$

when phase difference is ϕ, then

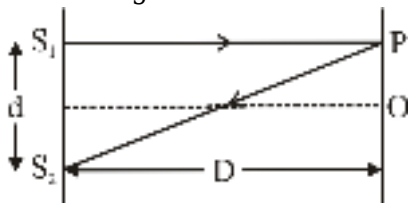
$I = I^0 + 4I^0 + 2\sqrt{4I_0^2}\cos\phi$

$= I_0 + 4I_0(1 + \cos\phi)$

$= I^0\left[1 + 8\cos^2\left(\frac{\phi}{2}\right)\right]$

$I = \frac{I_m}{9}\left[1 + 8\cos^2\left(\frac{\phi}{2}\right)\right]$

4) When dark fringe is obtained at the point opposite to one of the slits then



$S_1P = D$

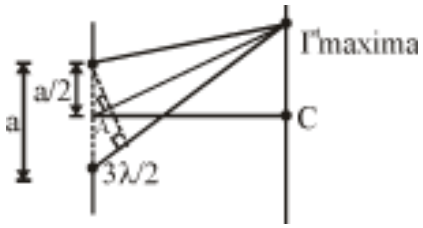
and $S_2P = \sqrt{D^2 + d^2} = D\left(1 + \frac{d^2}{D^2}\right)^{1/2} = D\left(1 + \frac{d^2}{2D^2}\right)$ Now, $S_2P - S_1P = D\left(1 + \frac{d^2}{2D^2}\right) - D = \frac{d^2}{2D}$

For dark fringe $S_2P - S_1P = \frac{(2n-1)\lambda}{2}$

So $\frac{d^2}{2D} = \frac{(2n-1)\lambda}{2}$

So $\lambda \propto d^2$

5)



For 1st maxima, path difference in wavelets from edges = $\frac{3\lambda}{2}$

 path difference in between wavelets from one of the edge and from then centre of slit = $\frac{3\lambda}{4}$

phase difference = $\frac{3}{4} \times 2\pi = \frac{3\pi}{2}$ radian.

6)

In diffraction of light at a single slit, the pattern obtained consists of a central bright band having alternate dark and bright bands of decreasing intensity on both sides.

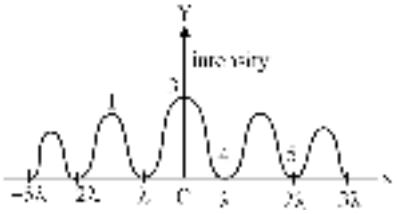
For n^{th} minima (Secondary)

$\sin\theta_n = \frac{n\lambda}{a}$

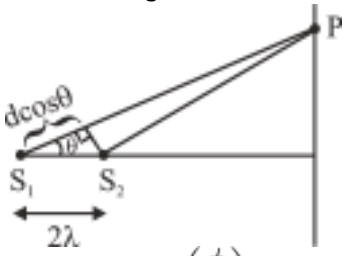
for n^{th} maxima (secondary)

$\sin\theta_n = (2n+1)\frac{\lambda}{2a}$

Thus point 5 corresponds to the path difference of extreme rays equal to 2λ



7) In Young's double slit experiment, intensity at a point is given by



$I = I_0\cos^2\left(\frac{\phi}{2}\right)$

But $\frac{I}{I_0} = \frac{3}{4}$ (given) or $\cos^2\left(\frac{\phi}{2}\right) = \frac{3}{4}$

or $\cos\frac{\phi}{2} = \frac{\sqrt{3}}{2}$ or $\phi = 60 = \frac{\pi}{3}$

Phase difference, $\phi = \frac{2\pi}{\lambda}$ × path difference

From the figure, path difference is

$d\cos\theta = 2\lambda\cos\theta \quad (d = 2\lambda)$

$\therefore \frac{\pi}{3} = \frac{2\pi}{\lambda} 2\lambda\cos\theta \quad \therefore \cos\theta = \frac{1}{12}$

$\therefore \theta = \cos^{-1}\left(\frac{1}{12}\right)$

8) Nuclear forces have the following well-known properties:

• They act over a very **short range** (≈1-2 fm).

• They are **attractive** at intermediate distances (responsible for blinding the nucleus).

• They are **charge independent,** i.e., the force between proton-proton, neutron-neutron, and proton-neutron pairs is nearly the same.

9) $P = \frac{NE}{t}$

$\frac{N}{t} = \frac{P}{E} = \frac{10^3}{200 \times 10^6 \times 1.6 \times 10^{-19}}$

$= 3.125 \times 10^{13}$

10) According to Rutherford atomic model, most of mass of atom and all its positive charge is concentrated in tiny nucleus & electron revolve around it.

According to Thomson atomic model, atom is spherical cloud of positive charge with electron embedded in it.

Hence,

Statement I is true but statement II false.

11)

For doubly ionised lithium atom -

$\frac{1}{\lambda_1} = R(3)^2\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = 9R\left(\frac{3}{4}\right) = \frac{27R}{4}$

For deuterium

$\frac{1}{\lambda_2} = R(1)^2\left[\frac{1}{1^2} - \frac{1}{(2)^2}\right] = \frac{3R}{4}$

$\frac{\lambda_1}{\lambda_2} = \frac{1}{9}$ i.e. 1 : 9

12)

Z' = Z – 2n+n'

A' = A – 4n

13) $n = \frac{P\lambda}{hc}$

14)

Solution :

Power of lamp $= 100\,\text{W} = 100\,\text{J s}^{-1}$.

Only 5% appears as visible light, so useful power:

$$P = 0.05 \times 100 = 5\,\text{J s}^{-1}$$

Wavelength of visible light:

$$\lambda = 5.6 \times 10^{-5}\,\text{cm} = 5.6 \times 10^{-7}\,\text{m}$$

Energy of one photon:

$$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^{8}}{5.6 \times 10^{-7}} \approx 3.55 \times 10^{-19}\,\text{J}$$

Number of photons emitted per second:

$$N = \frac{P}{E} = \frac{5}{3.55 \times 10^{-19}} \approx 1.4 \times 10^{19}$$

15)

Formula :
eV0 = h f_max - φ
$f = \frac{\omega}{2\pi}$

Solution : Angular frequencies present: 7ω, 10ω, 15ω
Maximum angular frequency = 15ω
So, $f_{max} = \frac{15\omega}{2\pi}$
Now, $V0 = \left(\frac{h}{e}\right) \times \left(\frac{15\omega}{2\pi}\right) - \frac{\varphi}{e}$

Final Answer : option (2)

16) $E = \frac{V}{d}$
V = E.d
V = 4.1 = 4 volt
$eV_0 = E - f$
$4eV = \frac{12400}{2000} - \phi$
$\phi = 6.2 - 4 = 2.2$ eV

17) $\lambda = \frac{h}{p}\left[\lambda = 7.5 \times 10^{-12}\text{m}\right]$

$$p = \frac{h}{\lambda}$$

$$\text{K.E.} = \frac{P^2}{2m} = \frac{\left(\frac{h}{\lambda}\right)^2}{2m} = \frac{\left[\frac{6.6\times10^{-34}}{7.5\times10^{-12}}\right]^2}{2 \times 9.1 \times 10^{-31}}$$

$$\text{K.E.} = \frac{4.195 \times 10^{-15}\text{J}}{1.6 \times 10^{-19}}\text{eV}$$

KE ≈ 26.18 Kev

This value is close to 25 KeV

18) $$\frac{\lambda_\alpha}{\lambda_d} = \frac{\frac{h}{m_\alpha v_\alpha}}{\frac{h}{m_d V_d}} = \frac{m_d v_d}{m_\alpha v_\alpha} = \frac{(m_d)(2v)}{(2m_d)(v)} = 1$$

The Ratio is 1 : 1

19) $$I = \frac{P}{4\pi d^2} = \frac{nhv}{A}$$

$$n = \frac{PA}{4\pi h d^2 v}$$

$$\frac{PA}{4\pi h} = \text{cons tan t(K)}$$

$$\therefore n = \frac{k}{d^2 v}$$

for 1st case

$$x = \frac{k}{d_1^2, v_1} \text{.....(1)}$$

And, for 2nd case

$$x' = \frac{k}{d_2^2 v_2} = \frac{x d_1^2 v_1}{d_2^2 v_2}$$ (from equation (1))

$$x' = \frac{x(2)^2 \times 10^{14}}{(4)^2 \times 2 \times 10^{14}} = \frac{x}{8}$$

$$\boxed{x' = \frac{x}{8}}$$

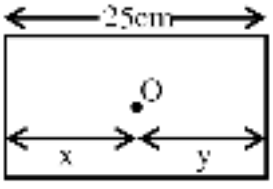
20)

$m_1 = - m_2$

$$\frac{f}{f+5} = -\frac{f}{f+15}$$

2f = – 20

f = –10 cm

21) 

$\frac{x}{\mu} = 6 \Rightarrow \frac{y}{\mu} = 4$

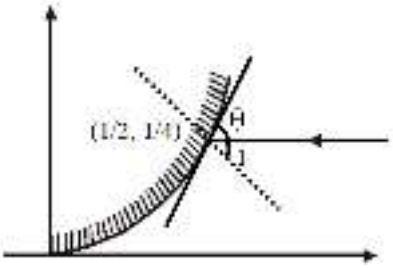
x + y = 25 cm

6μ + 4μ = 25

μ = 2.5

22)

Variable refractive index causes looming and mirage.

23) 

$\tan\theta = \frac{dy}{dx} = 2x$

$\frac{dy}{dx} = 2\left(\frac{1}{2}\right) = 1$

θ = 45°

i = 45° ⇒ r = 45°

24) $m = \frac{f}{u + f}$

$m' = \frac{f}{-x + f}$, $m = \frac{f}{-y + f}$

⇒ m' = –m

⇒ $\frac{f}{-y + f} = \frac{-f}{-x + f}$ □ y -f = f -x

⇒ $f = \left(\frac{x + y}{2}\right)$

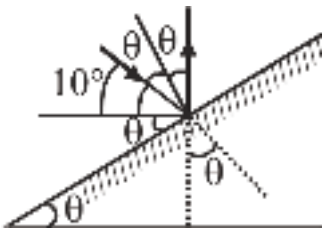
25) $L = V_0 + |u_e|$

$= \frac{f_0 u_0}{f_0 + u_0} + f_e \Rightarrow \frac{1 \times (-1.1)}{1 - 1.1} + 5$

= 11 + 5 = 16 cm

26)

θ + 10 + θ = 90



θ = 40°

27) $\mu_1 = \mu_3$ since there is no bending at first surface. $\mu_3 < \mu_2$ since the ray bends towards normal as it passes from μ_3 to μ_2 medium.

28) u = +20, m = –1

$$m = \frac{f}{f-u}$$

29) We know that $P_{eq} = \Sigma P_i$

 given all lenses are identical

 $5P = 25D \Rightarrow P = 5D$

$$\therefore \frac{1}{f} = 5 \Rightarrow f = \frac{1}{5}m = 20cm$$

30) For lens 1 : $f_1 = 10$, u = –30, v = ?

$$v = \frac{uf}{u+f} = \frac{-30 \times 10}{-30+10} = 15$$

For lens 2 : $f_1 = -10$, u = 10, v = ?

$$v = \frac{uf}{u+f} = \frac{10 \times -10}{10-10} = \infty$$

For lens 3 : f = 30, u = –∞, v = ?

So v will be 30.

31) As $\lambda_{red} > \lambda_{yellow} > \lambda_{violet}$

Light ray with longer wavelength bends less.

32) $$\frac{\mu_A}{\mu_B} = \frac{c/V_A}{c/V_B} = \frac{V_B}{V_A} = \frac{1}{2}$$

Let the thickness is d

$$\frac{d}{v_B} - \frac{d}{v_A} = 5 \times 10^{-10} \Rightarrow d = \frac{5 \times 10^{-10} \times v_A v_B}{v_A - v_B}$$

As $v_A = 2v_B \Rightarrow d = 5 \times 10^{-10} \times 2v_B$

Or $d = 5 \times 10^{-10} \times v_A$

33) $$v = \frac{uf}{u+f}$$

Case-I

If v = u

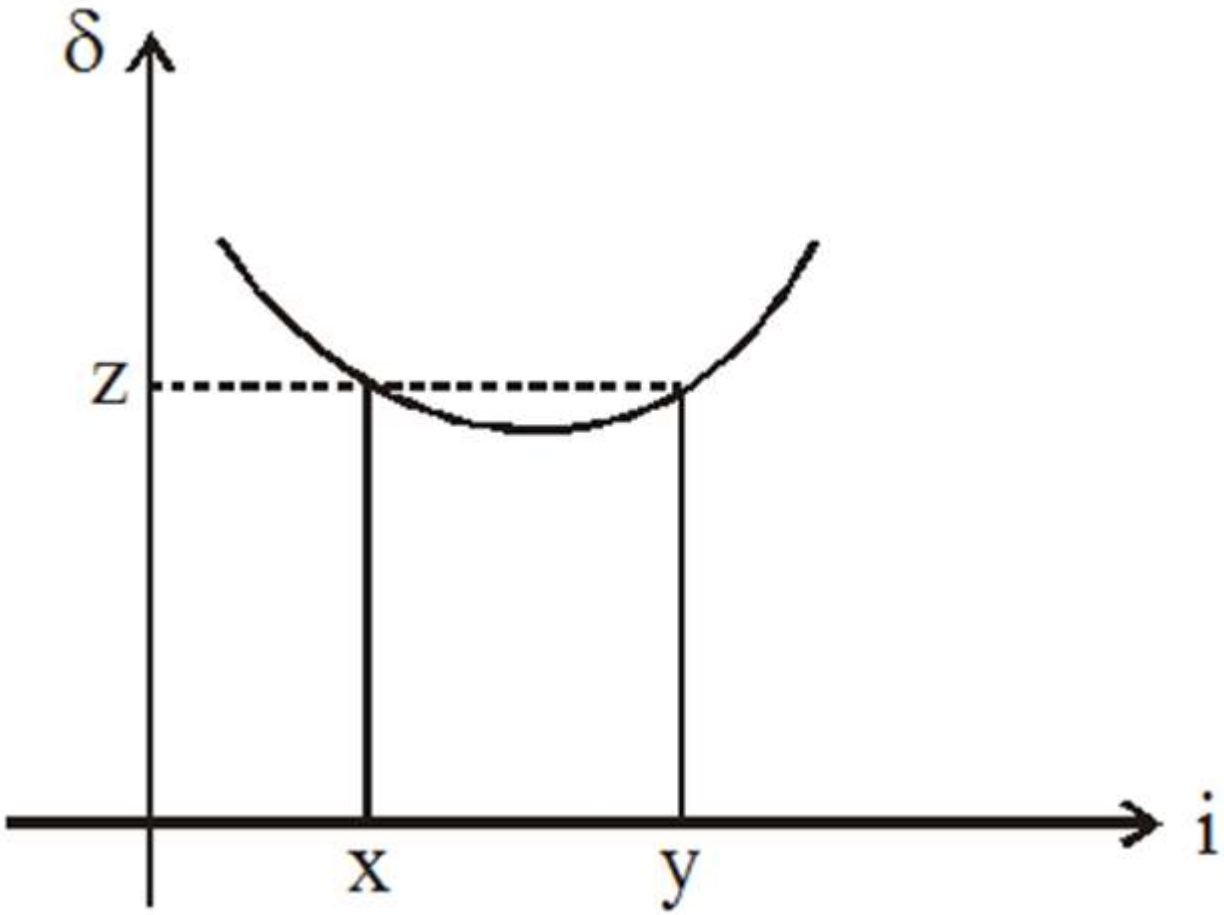
$\Rightarrow$ f + u = f $\Rightarrow$ u = 0

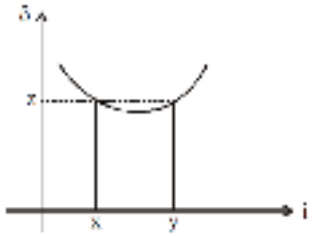
Case-II

If u = ∞ $\Rightarrow$ v = f

Alternate solution :
For a concave lens, the object distance u, image distance v, and focal length *f* are negative according to the Cartesian sign convention. The relation between u and v follows the lens formula and represents a rectangular hyperbola. The image formed is always virtual, erect, and diminished, hence its distance from the lens is less than that of the object ($|v| < |u|$). Therefore, in the u-v graph, the curve lies below the reference line u = v. As the object moves far away, the image approaches the focus, so the curve asymptotically approaches the line v = *f*.

34)





Deviation by prism
δ = i + e – A
Using graph
At i = x, e = y, δ = z
Put values,
Z = x + y – A
Therefore A = x + y – Z

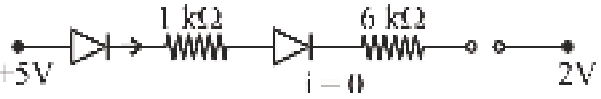
35)

Direction of diffusion current is P to N and direction of drift current is N to P.

36)

Both diode is in F.B. but due to less potential barrier of Ge diode, current will pass only from Ge diode.

$\Rightarrow I = \frac{12-0.3}{5\,k\Omega} = 2.34\ mA$

37) 

38) For zener brakdown supply voltage > breakdown voltage

39) $I_L = \frac{V_Z}{R_L} = \frac{6}{4\,k\Omega} = 1.5\,mA$

for maximum Zener current $V_{in} = 16V$

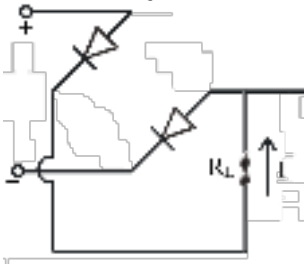
$I = \frac{V - V_Z}{R_S} = \frac{16-6}{2\,k\Omega} = 5\,mA$

$(I_Z)_{max} = I - I_L = 5 - 1.5 = 3.5$ mA

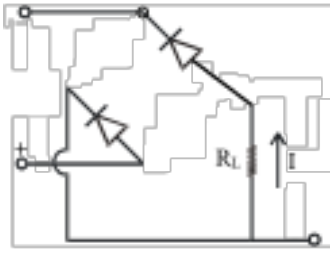
40)

In given option-4 work as full wave rectifier.

For + Ve cycle



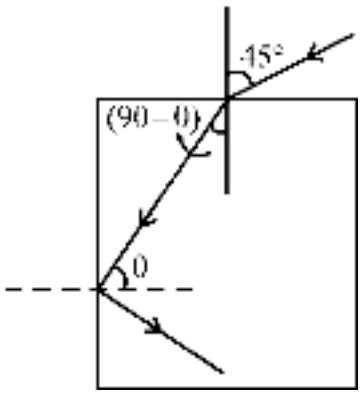
For – Ve cycle



41) **Explain the Question :** To find the minimum refractive index of slab so that light will refract back.

Concept : This question is based on critical angle and total internal reflection.

Solution :



$1 \times \sin 45° = \mu \times \sin(90 - \theta)$

$\frac{1}{\sqrt{2}} = \mu \cos\theta \Rightarrow \cos\theta = \frac{1}{\sqrt{2}\mu}$

$\sin\theta \geq \frac{1}{\mu}$

$\Rightarrow \sqrt{1-\cos^2\theta} \geqslant \frac{1}{\mu}$

$\Rightarrow \sqrt{1-\frac{1}{2\mu^2}} \geqslant \frac{1}{\mu} \Rightarrow \mu \geq \sqrt{\frac{3}{2}}$

Final Answer : 3

42) In reflected ray polarization becomes perpendicular to the plane of incident. Here in the incident ray there is no component perpendicular to the plane hence there will not be any reflected ray.

43) $\lambda_{min} = 400$ nm $\quad \lambda_{max} = 600$ nm

$E_{max} \frac{1240}{400} = 3.1$ eV $\quad E_{min} \frac{1240}{600} = 2.06$ eV

$\phi = 2.39$ eV

$\lambda E_{max} = 3.1 - 2.39 = 0.71$ eV

$R = \frac{mV}{qB} = \frac{\sqrt{2mK}}{qB} \Rightarrow B = \frac{\sqrt{2mK}}{qR}$

$B = \frac{1}{R} = \sqrt{\frac{2mK}{C^2}}$

$= \frac{1}{R} = \sqrt{\frac{2 \times 9.1 \times 10^{-31} \times 0.71 \times e}{e^2}}$

$= \frac{1}{R} = \sqrt{\frac{2 \times 9.1 \times 10^{-31} \times 0.71 \times 1.6 \times 10^{-19}}{1.6 \times 10^{-4}}}$

$\Rightarrow B = 2.85 \times 10^{-5}$ T

44) $E_g = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{660 \times 10^{-9}}$J

$= \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{660 \times 10^{-9} \times 1.6 \times 10^{-19}}$eV

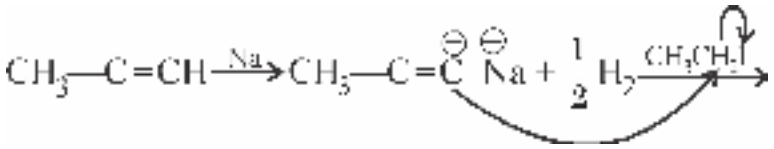
$= \frac{15}{8}$eV

So x = 15

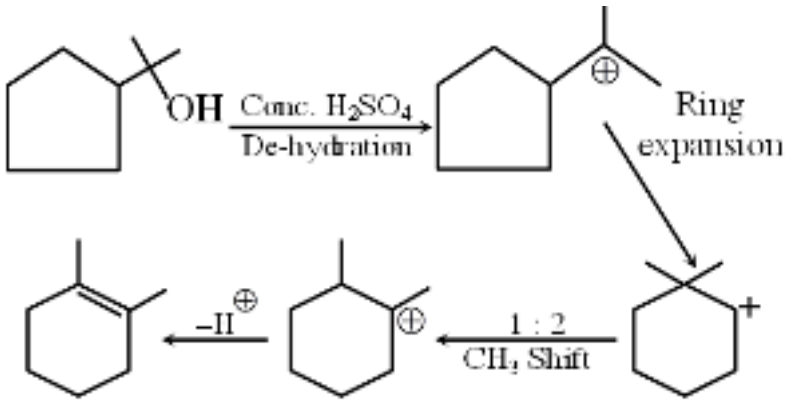
45)

u = – (55.0 – 25.0) = –30.0 cm
v = –(85.0 – 25.0) = –60.0 cm
$f = \frac{uv}{u+f} = \frac{(-30.0)(-60.0)}{-30.0-60.0}$ = – 20. cm
$\Rightarrow \Delta f = \pm\left(\frac{\Delta u}{u^2} + \frac{\Delta v}{v^2}\right) f^2$
Also Δu = Δv = ±0.2
$\Rightarrow \Delta f = \pm\left(\frac{0.2}{900} + \frac{0.2}{3600}\right) 400$
$= \frac{0.2}{3600} \times 5 \times 400 = \pm 0.11$
Δf = ± 0.1 cm ⇒ f = –20.0 cm ± 0.1 cm

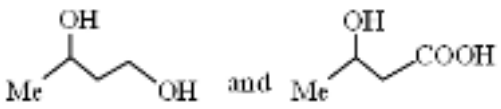
CHEMISTRY



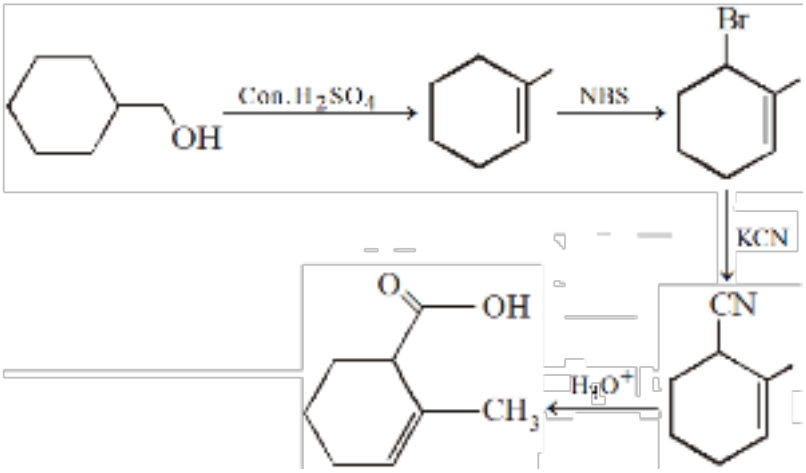
46)
CH_3–C≡C–CH_2CH_3



47)

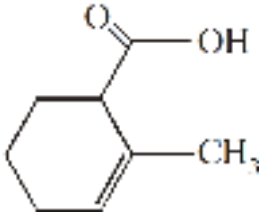


48)
$NaBH_4$ is a weak reducing reagent. So it reduces only carbonyl group while reduction of –COOH group does not happen and it remains unaffected.
$LiAlH_4$ is strong reducing reagent; so it can reduce both carbonyl and carboxylic groups.



49)

Final Answer :

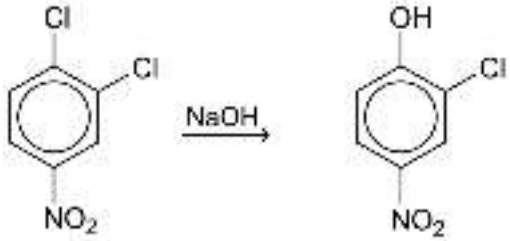


50)

Solution:

In S_NAr reactions, strong electron-withdrawing groups like $-NO_2$ activate only the ortho and para positions by stabilizing the Meisenheimer complex.

Hence, only the para-Cl (resonance-activated) is replaced by OH^-, while the meta-Cl remains unchanged.



Final Answer : Option(1)

51) Comp. containing $CH_3-\underset{\underset{OH}{|}}{CH}-H/R$ group shows haloform or iodoform test.

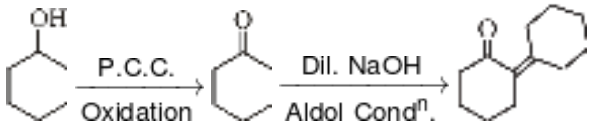
52) $Ph-\overset{\overset{O}{\|}}{C}-Cl \xrightarrow[Pd\text{-}BaSO_4]{H_2}$ Aldehyde (Rosenmund Reduction)

$Ph-C\equiv N+SnCl_2-HCl \rightarrow (A) \xrightarrow{H_3O^{\oplus}}$ Aldehyde (Stephen reduction)

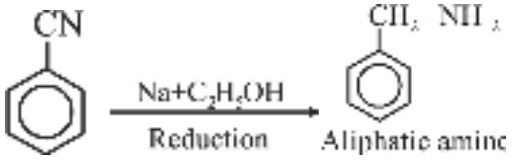
$Ph-C\equiv N \xrightarrow[(2)\ H_2O/H^{\oplus}]{(1)\ RMgX} Ph-\overset{\overset{O}{\|}}{C}-R$

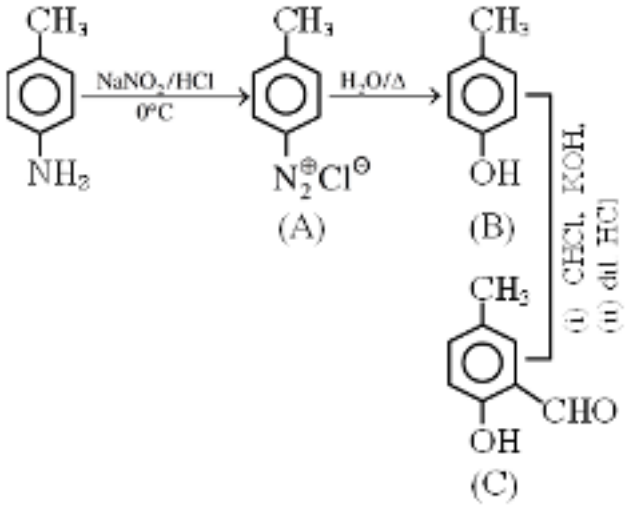
$Et-\overset{\overset{O}{\|}}{C}-O-Et \xrightarrow[H_2O]{DIBAL\text{-}H}$ Aldehyde

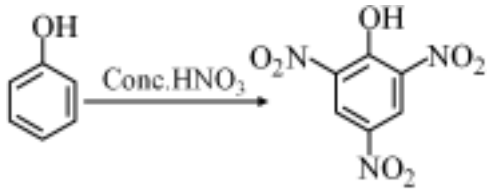
Final Answer: Option (3)

53) 

54) LAH reduces –COOH to $–CH_2OH$.

55) 

56) 

57) 

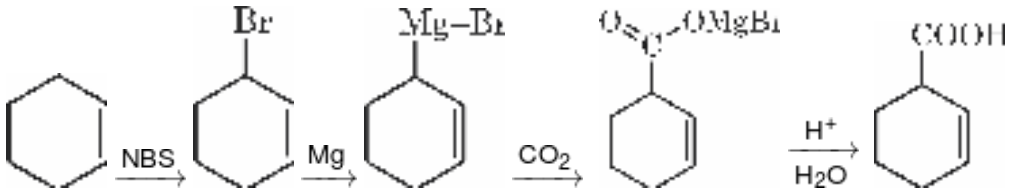
58) Theory Based

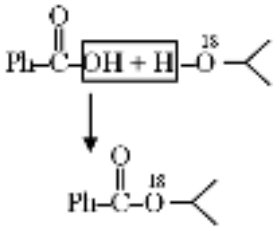
59) Amylose is the linear polymer of α-D glucose formed by 1 : 4 glycosidic linkage.

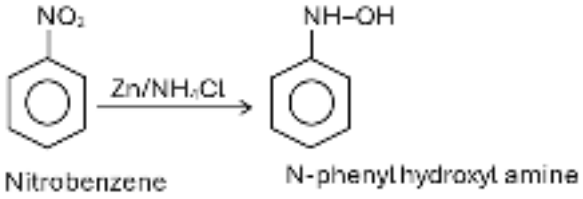
60) Reducing sugars will give positive tollen’s test.

61)

Hydrazine is not an organic compound so can't shows lassaigne test.

62) 

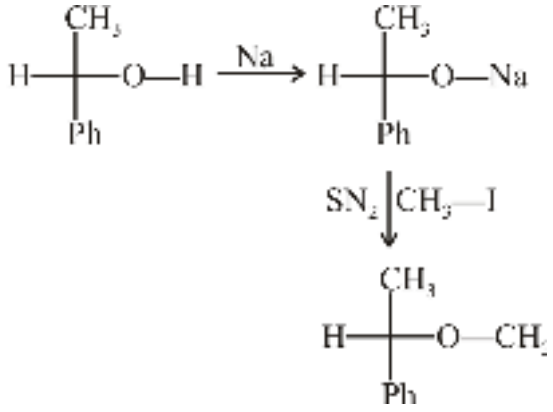
63) 

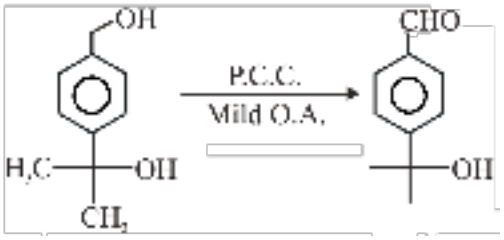
64) 

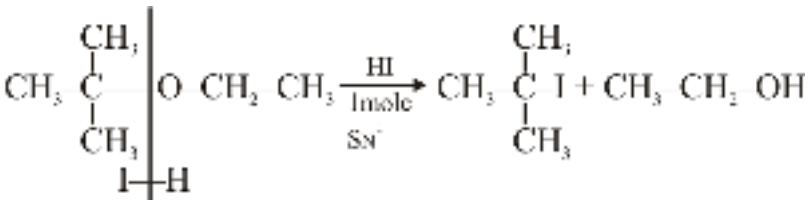
Conclusion: Hence, option (4) is correct.

65) Rate of NAR ∝ +ve charge on carbonyl carbon

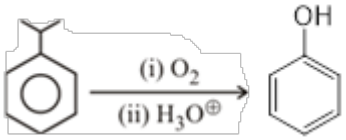
$$\propto \frac{1}{\text{Steric hindrance}}$$

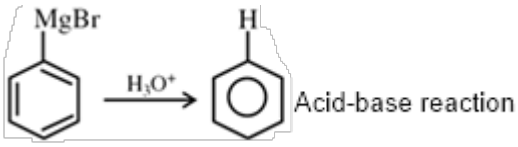
66) 

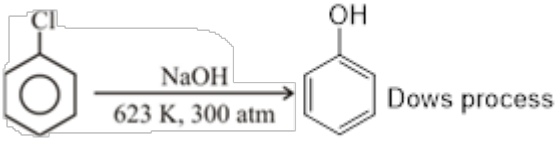
67) 

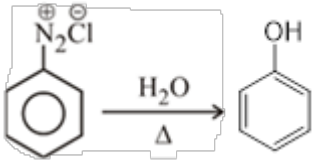
68) 

69) **Given reaction**

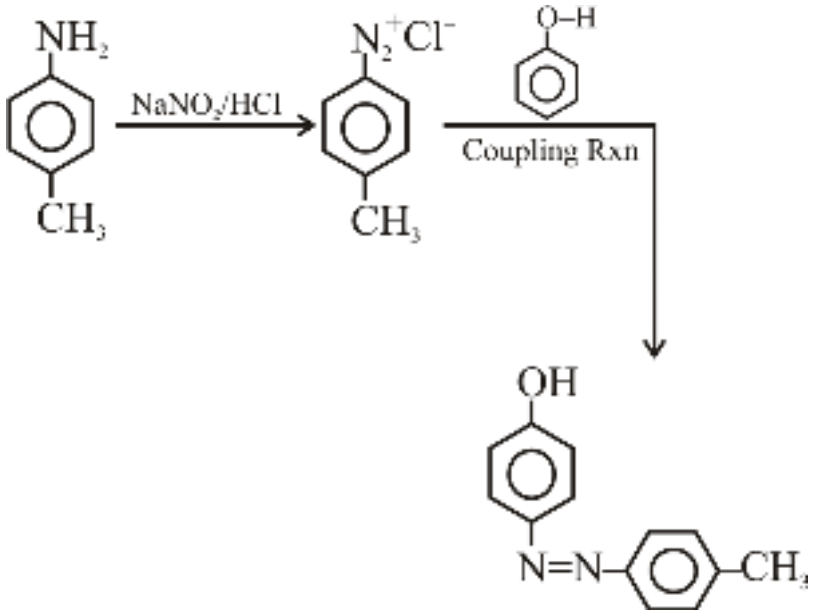
1. 

2. 

3. 

4. 

Final answer: option (2)

70) 

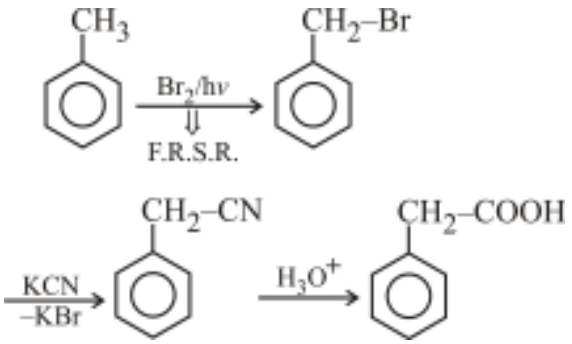
71)

This is Hofmann bromanide reaction and $RCONBr_2$ does not form in this reaction. See mechanism for it :

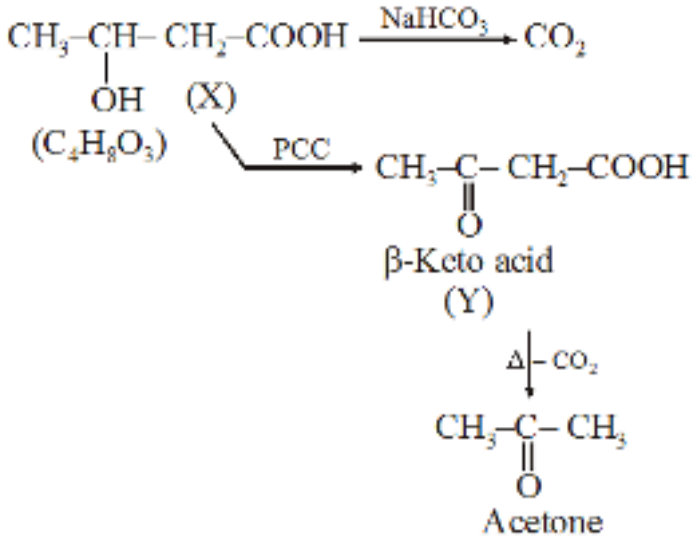
$RCO–NH_2 + Br_2 + 4KOH \rightarrow R–NH_2 + K_2CO_3 + 2KBr + 2H_2O$

$R–CONBr_2$ does not form as intermediate

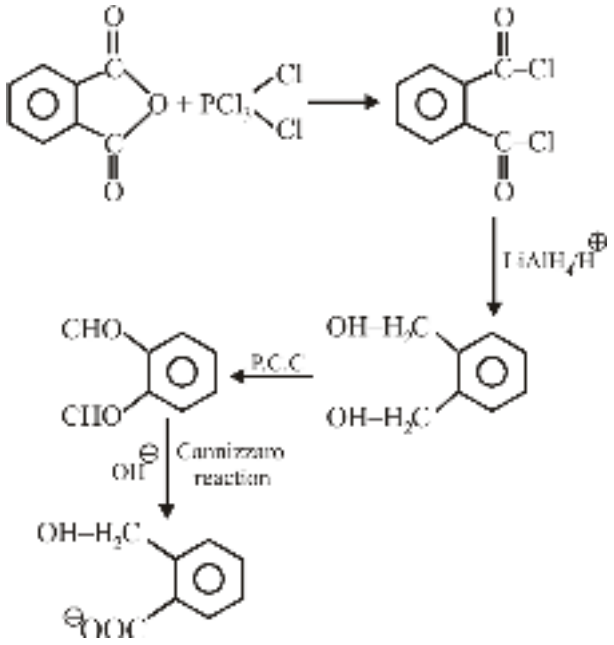
72)



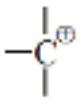
73)



74)



75) Reactivity for N.S.R ⇒ Acid Halide > Anhydride > Ester > Amide

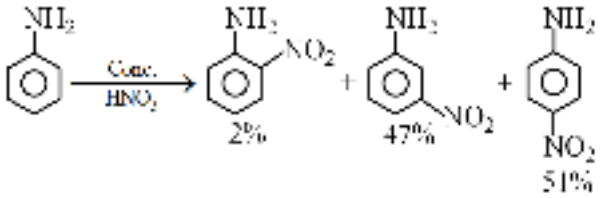
76) Reactivity of SN_1, $R\propto^n$ ∝ stability of 

77) Amylopectin is a homopolymer of a D-glucose having $C_1 - C_4$ - glycocidic linkage (for linear chain formation) and $C_1 - C_6$ linkage (for branch chain formation).

78) **Solution:**
• Fehling's Test: Acetaldehyde, being an aliphatic aldehyde, reduces Fehling's solution to give a brick-red precipitate of Cu_2O. Benzaldehyde, an aromatic aldehyde, does not react with Fehling's solution due to the absence of alpha-hydrogens and its lower reactivity.
• Iodoform Test: Acetaldehyde contains a methyl keto group (CH_3CO-) and thus reacts with I_2/NaOH to form a yellow precipitate of iodoform (CHI_3). Benzaldehyde lacks this methyl group and gives a negative result.
• Tollens' Reagent: Both compounds are aldehydes and will reduce Tollens' reagent to form a silver mirror; therefore, it cannot be used to differentiate them.

Final Answer: Option (4)

79)

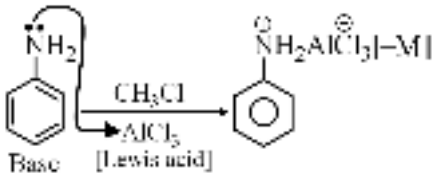


80) –M increase reactivity

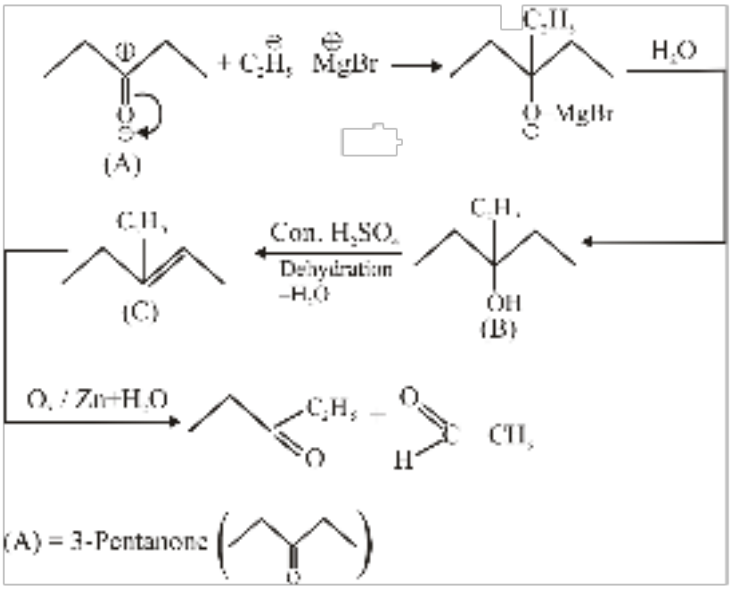
81) By $NaNO_2$ + HCl at 0°–5°C introduce $\overset{\oplus}{N_2}\overset{\ominus}{Cl}$.

82) $C_6H_6 \rightarrow C_6H_5$–$NO_2 \rightarrow C_6H_5NH_2$

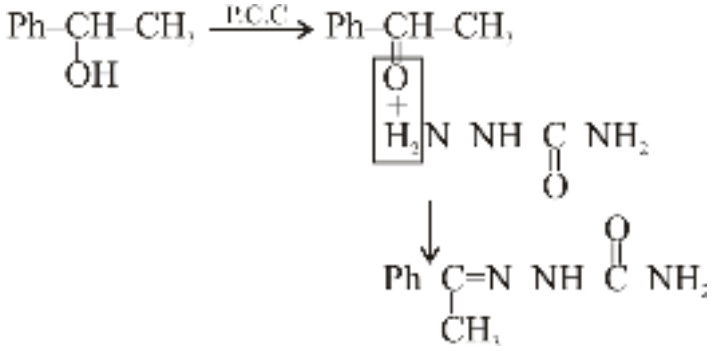
83) ⇒ If $\overset{-I}{\overline{-M}}$ group attached directly to Benzene then it do not give Friedel craft reaction.
⇒ ANILINE also does not give Friedel craft reaction because Aniline itself is a base and reagent used in FCR $AlCl_3$ is Lewis Acid.
 Acid-Base reaction takes place and converts -NH_2 group of Aniline to deactivating group [-M] and hence Aniline do not give FCR.



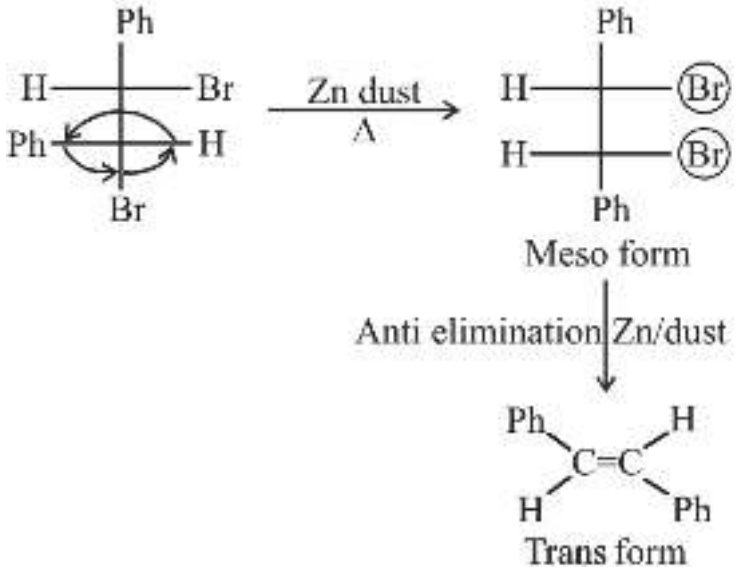
84)



85)



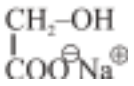
86)

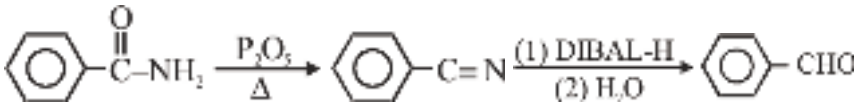


Meso upon Anti elimination gives trans isomer as major product

87) A → 

B → 

C → 

88) 

89) 1 Maltose structure
2 Lactose structure
3 Sucrose structure
4 Fructose structure

90) Kolbe reaction followed by esterification.

BIOLOGY

91) NCERT Pg. # 209 & 213

92) NCERT Pg. # 212, 213

93) NCERT Pg. # 207

94) NCERT Pg. # 209

95) NCERT Pg. 207

96) NCERT Pg. # 211

97) NCERT Pg. # 209

98) NCERT, Pg. # 212

99)

NCERT XII Pg.# 210

100) NCERT, Pg. # 209

101) NCERT-XII, Pg. # 214

102) NCERT Pg. # 209, 224, 225

103) NCERT Pg. # 222,223

104) NCERT Pg. # 224-225

105) NCERT Pg.# 225

106)

NCERT Pg.# 201

107)

NCERT Pg.# 201

108)

NCERT Pg.# 200-201

109)

NCERT Pg.# 200

110)

NCERT-XII Pg. # 242 [E], 266 [H]

111)

NCERT-XII, Pg. # 262

112) NCERT-XII, Pg # 20, 30, 192, 202

113)

NCERT Pg.# 200, 201

114)

NCERT Pg.# 197

115)

NCERT Pg. # 198

116) NCERT, Pg. # 201

117) NCERT Pg. # 200

118) NCERT, Pg. # 199

119)

Module 10

120) NCERT Pg. # 194

121)

NCERT Pg. # 191

122) NCERT Pg. No. # 192

123) NCERT Pg. No. # 193

124) NCERT, Pg. # 196

125) NCERT Pg. # 194

126) NCERT Pg.# 227, 231, 232, 229(E); 249, 250, 251, 252(H),

127)

NCERT Pg. # 210

128) XII NCERT Page No. # 267

129) NCERT XII Pg.# 219

130)

NCERT-XII Pg.# 219

131)

NCERT XII Pg.# 221

132)

NCERT XII Pg.# 221

133) NCERT XII Pg.# 221

134)

NCERT XII Pg. No :- 266

135)

NCERT XII Pg.# 224-225

136)

NCERT, Pg. # 119

137) NCERT Pg # 118, 119

138)

NCERT, Pg. # 115

139)

NCERT, Pg. # 111

140)

NCERT, Pg. # 116

141) NCERT Pg. # 120,121

142) NCERT Pg # 111

143) NCERT Pg # 111

144) NCERT- XII Pg.# 120

145) NCERT Pg # 121

146)

NCERT, Pg. # 121

147) NCERT-XII, Pg # 122

148) NCERT Pg # 113

149)

NCERT, Pg. # 111

150) NCERT Pg.#153

151) NCERT Pg # 136

152)

Correct Answer: (1)

153)

NCERT, Pg. # 134, 135

154)

NCERT, Pg. # 142, 143

155) NCERT-XII, Pg. # 134 to 136

156) NCERT XII Pg. No. # 143

157)

NCERT, Pg. # 130

158)

NCERT, Pg. # 135

159) NCERT (XII) Page # 132/145(H) Para: 7.1

160) NCERT, Pg. # 140

161) NCERT, Pg. # 136

162) NCERT (XII) Page#137/150(H) Para: 7.2.5

163) NCERT Pg. # 134, 135

164)

NCERT XII Page No. # 130(E)/142(H)

165) NCERT_Pg. # 135-174

166) NCERT, Pg. # 133

167)

NCERT, Pg. # 138

168) NCERT XII Pg. No. # 133

169) NCERT XII Pg. No. # 143, 145

170)

NCERT, Pg. # 141

171)

NCERT, Pg. # 143

172)

NCERT, Pg. # 144

173) NCERT-XII, Pg. # 146

174)

NCERT, Pg. # 142

175)

NCERT, Pg. # 140

176)

NCERT, Pg. # 132

177) NCERT, Pg. # 132

178) NCERT XII Pg. No. # 131

179)

NCERT, Pg. # 131

180) NCERT (XII) Page#135-136/147-149(H) Para: 7.2.2